



MSCC Operator's Guide — Windows (Preliminary)

Multus SDR Control Console for Proficio, Geminus and Ultimus transceivers

Instructions based on MSCC NET9

This preliminary edition covers Windows local operation and firmware updates (Appendix A).

Coming: Appendix B: The Electronic Keyer · Appendix C: Installing and Upgrading · Appendix D: Where Settings and Calibration Files Live · Appendix E: Troubleshooting (FAQ) · Appendix F: WSJT-X and fldigi Setup.

Separate guides for Linux / Raspberry Pi and for remote operation are coming.

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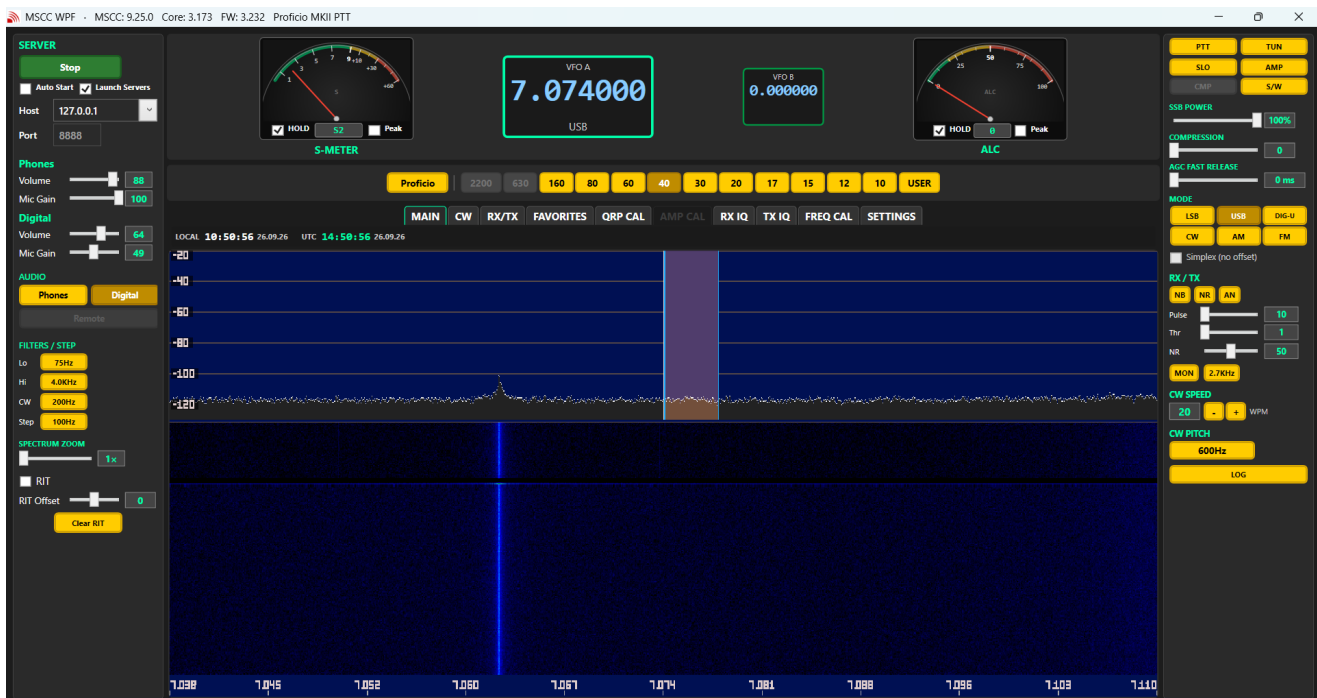
□ WELCOME

This guide covers installing and using MSCC, the Multus SDR Control Console. MSCC is the control program for Multus SDR, LLC transceivers: Proficio, Geminus and Ultimus, in both Legacy and MKII versions. It works only with Multus SDR equipment.

MSCC comes in two parts:

- **The servers.** These small programs talk to the radio over USB and do the receive and transmit signal processing. They run on the computer the radio is plugged into. In this guide, that is your Windows PC.
- **The client.** This is the window you operate from. The Windows client is called **MSCC for Windows**. These instructions are based on MSCC NET9.

The client and servers talk over your network. The client can run on the same computer as the radio (“local”). It can also run on another computer in the house or across the internet (“remote”). Remote operation has its own guide.



MSCC for Windows main window, MAIN tab showing



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☐ 1. Overview (READ THIS FIRST!)

Please read this section before you install anything.

1. **Firmware first.** The radio's firmware should match this version of MSCC. The window title shows the firmware version (FW) after you connect. Firmware updates are covered in Appendix A.
2. **Your setup.** This guide is for *Windows local*: the radio plugs into a Windows PC, and MSCC for Windows starts the servers for you. See Section 2. For a Raspberry Pi or Linux PC, or for remote operation, see the separate guides.
3. **Calibrate once.** A new radio or new installation should get frequency calibration (FREQ CAL), power calibration (QRP CAL) and transmit image balance (TX IQ). Factory values are loaded automatically for your radio model, so this is a touch-up, not a must-do on day one.
4. **Bug reports and feature requests:** post them on our forum, <https://multusdr.groups.io/g/main>



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□ 2. Windows Local Operation (Instructions based on MSCC NET9)

2.1 Installing and starting for the first time

1. **Install the Microsoft .NET 9 runtime (first-time install only).** MSCC needs this free Microsoft software to run.

- Go to <https://dotnet.microsoft.com/en-us/download/dotnet/9.0>.
- Find the heading **.NET Desktop Runtime 9.0** (the number after 9.0 changes as Microsoft releases updates; any 9.0 version is fine). In its **Windows** row, under **Installers**, download and run **both** the **x64** and the **x86** installer (boxed in red in the picture below). MSCC may not start unless both are installed.
- The Desktop Runtime already includes the plain “.NET Runtime”, so you don’t need to install that separately.
- You only need to do this once per computer. If a runtime is already installed, its installer will offer **Repair** or **Uninstall** instead of installing; just close it.

.NET Desktop Runtime 9.0.20

The .NET Desktop Runtime enables you to run existing Windows desktop applications. **This release includes the .NET Runtime; you don't need to install it separately.**

*

OS	Installers	Binaries
Windows	x64 x86 Arm64 winget instructions	x64 x86 Arm64

.NET Runtime 9.0.20

The .NET Runtime contains just the components needed to run a console app. Typically, you'd also install either the ASP.NET Core Runtime or .NET Desktop Runtime.

The .NET 9.0 download page. Under .NET Desktop Runtime 9.0, Windows row, download both the x64 and the x86 installer (boxed in red).

2. Run the newest installer you were given. Its name looks like `mscc-net9-R9-26-6-install.exe` (that is only an example; the number on yours will differ). It installs the MSCC for Windows client *and* the Windows servers. The usual install folder is **C:\mscc-net9**.

How to read the version name. Take `mscc-net9-R9-26-6` as an example. **net9** means it runs on Microsoft .NET 9. **R9-26-6** is the release: month 9 (September), day 26, and the 6th build made that day. A later date is newer. Always use the newest



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installer you were given. The window title bar shows the same number, written like 9.26.6.

3. Plug in the radio's USB cable and power the radio on. On a PC that has never had a Multus radio, install the USB driver with Zadig first (Appendix A.2).
4. Start **MSCC** from its desktop shortcut, **MSCC-NET9**.
 - The first time, MSCC copies a starter set of settings into your Windows profile. There is no separate "Initialize.bat" step anymore.
 - Windows asks whether MSCC may use the network. There will be several of these prompts, one for each server. You must click **Allow access** on every one.
5. Go to the **SETTINGS** tab (Section 2.12) and set:
 - **COM Port**: the serial port the servers use for CAT (for logging and digital programs).
 - **Audio → Operator (phones) Out and Mic**: the speaker or headphones and the microphone you want to use.
 - **Audio → Digital Out and Mic**: the virtual audio cable devices, if you run WSJT-X or fldigi.
6. Press **Start** on the left side of the window.

Only one copy of MSCC can run at a time. If you try to open a second one, you'll see "MSCC is already running".

The window remembers its size and position. Your frequencies, modes, filters and display settings are also remembered between sessions.

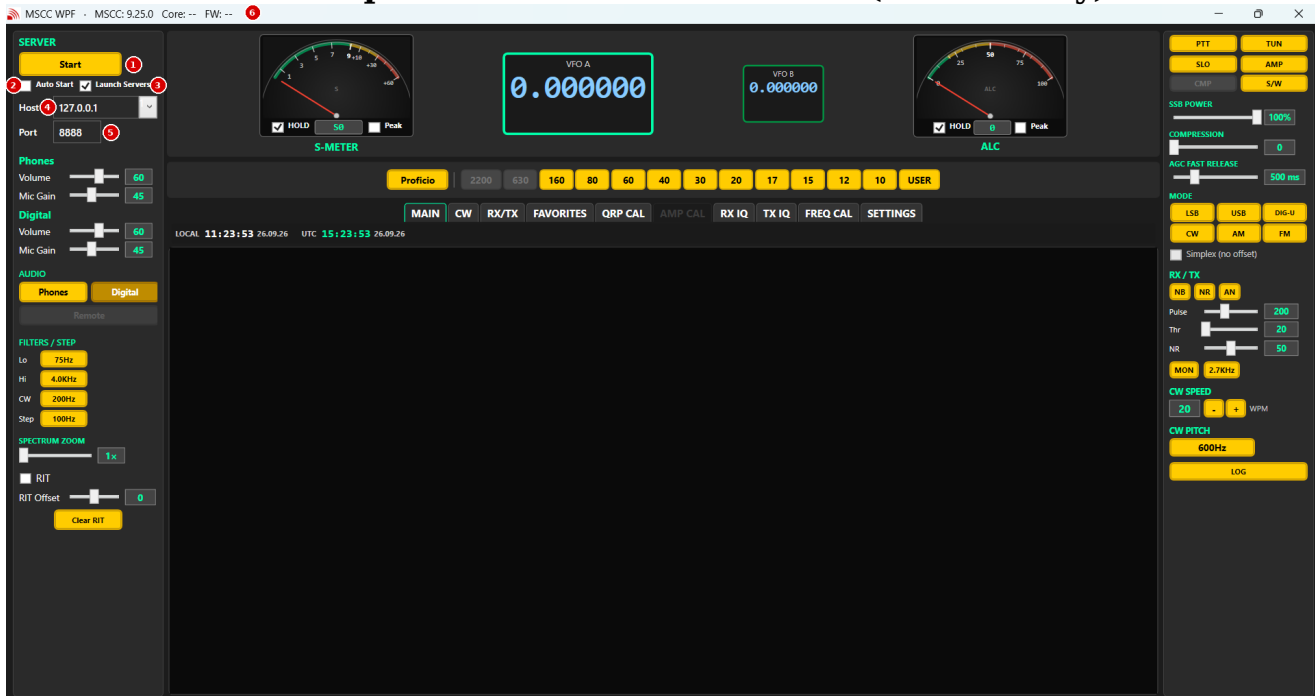
2.2 Connecting to the radio

On the left rail, under **SERVER**:

- **Launch Servers** checked (normal for Windows local): **Start** launches the servers on this PC, then connects.
- **Launch Servers** unchecked: **Start** only connects, to servers that are already running (on this PC or on another computer).
- **Host** is the address of the servers. For local operation use **127.0.0.1**. **After changing the IP address (or Port), close MSCC and restart it.** Stop and Start alone doesn't pick up the new address.
- **Port** is **8888** unless you changed it on the server.
- **Auto Start** presses Start for you whenever MSCC opens.



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MSCC after it opens, before Start is pressed

MSCC just after it opens, before you press Start: (1) **Start**, (2) **Auto Start**, (3) **Launch Servers**, (4) **Host**, (5) **Port**, (6) the title bar, which shows “Core: – FW: –” until you connect. The VFOs read 0.000000 and the spectrum stays empty until then.

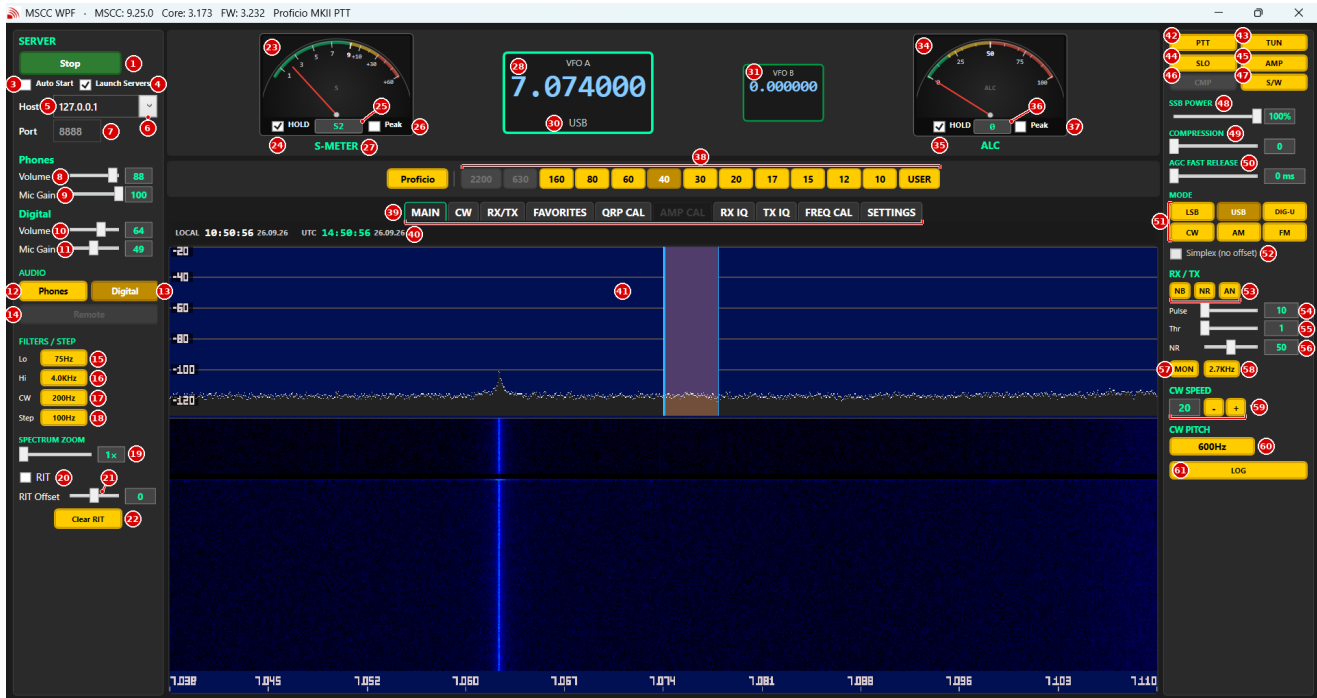
When connected, the Start button turns green and reads **Stop**. The title bar then shows the versions and your radio model. For example (the numbers on yours will differ): MSCC WPF · MSCC: 9.26.6 Core: ... FW: 3.232 Proficio MKII PTT

If something required is missing (COM port or operator audio), Start is grayed out. A yellow “Setup needed” line tells you what’s missing. Pressing Start anyway offers to open Settings for you.

Stop ends the session. It only closes the servers that this copy of MSCC started. Closing the MSCC window does the same.

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2.3 The Main Window



The main window with numbered callouts (overview; enlarged views follow with each list)

The numbers match the list below. Items 2, 29, 32 and 33 have no callout: they are blank in this screenshot (2 appears only when setup is incomplete, 29 and 32 only while RIT is in use, and 33 was empty because VFO B had not been used yet; first-time users won't have a VFO B yet).

The main window has three columns:

- **Left rail:** connection, audio, filters, zoom and RIT.
- **Center:** meters and VFOs at the top, the band buttons, then the tabs.
- **Right rail:** the operating controls (PTT, TUN, modes, noise controls, CW).

Both rails stay visible on every tab.

Button colors:

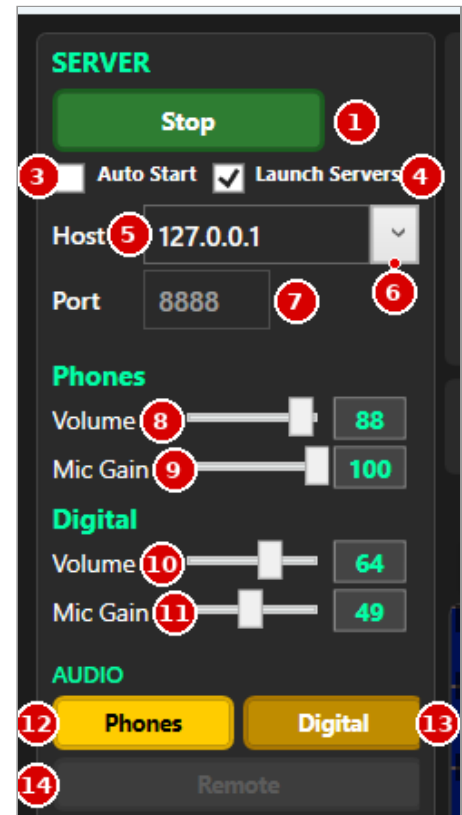
- Bright gold = off.
- Darker gold with white text = on or selected.
- **Red** = transmitting, or a transmit-related switch that is on (PTT, TUN, AMP, calibration TX).
- Gray = not available right now.

Mouse wheel: over any slider, one notch moves the slider one small step.

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Left rail

1. **Start / Stop** (SERVER). Connects to the radio and starts the servers if Launch Servers is checked. Green “Stop” while running. See 2.2.
2. **Setup status line** (yellow). Says what’s missing before you can Start. Hidden when all is well.
3. **Auto Start** checkbox. Starts automatically when MSCC opens.
4. **Launch Servers** checkbox. Checked: Start also runs the servers on this PC. Unchecked: connect only. Checking it turns Remote off.
5. **Host** box. Server address. Press Enter (or click elsewhere) to accept. **After changing the IP address, close MSCC and restart it.** Stop and Start is not enough; MSCC keeps using the old address until you restart the program.
6. **Recent hosts** (▼). Pick an address you used before.
7. **Port** box. Server port, normally 8888. Locked while connected.
8. **Phones Volume** (0–100). The radio’s headphone and speaker volume.
9. **Phones Mic Gain** (0–100). The radio’s microphone gain.
10. **Digital Volume** (0–100). Receive audio level sent to your digital program. Grayed while Remote is on.
11. **Digital Mic Gain** (0–100). Transmit audio level coming from your digital program. Grayed while Remote is on.
12. **Phones** button (AUDIO). Use your headset or speaker and microphone.
13. **Digital** button (AUDIO). Use the virtual audio cable for WSJT-X, fldigi and similar programs. Compression (CMP) turns off on this path.
 - Choosing the DIG-U mode switches to Digital for you. Leaving DIG-U puts back what you had.
 - If Remote is on, clicking Phones or Digital turns Remote off.



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14. **Remote** button. Makes *this* computer the operator position when the radio is somewhere else. It opens the Remote Audio window (2.14). It is grayed out when Launch Servers is checked or Host is 127.0.0.1, because the radio is local then. See Section 4.

15. **Lo** (low cut). Each click moves the low edge of the receive filter: 75 → 100 → 200 → 300 → 500 Hz, then back to 75.

16. **Hi** (high cut). Each click moves the high edge of the receive filter.

- In **DIG-U**: 1.0k → 1.4k → 2.4k → 2.7k → 3.0k → 4.0k → 5.5k, then back to 1.0k. **The 1.0k and 1.4k settings are for DIG-U only.** They are handy for narrow digital work.
- In **all other modes**: 2.4k → 2.7k → 3.0k → 4.0k → 5.5k.
- In AM the filter is $\pm$ the Hi value around the carrier. FM always uses at least ± 5.5 kHz.

17. **CW** (CW bandwidth). 200 Hz → 400 Hz → 1.8 kHz. Used in CW mode.

- Lo, Hi and CW are remembered separately for each mode and each band.

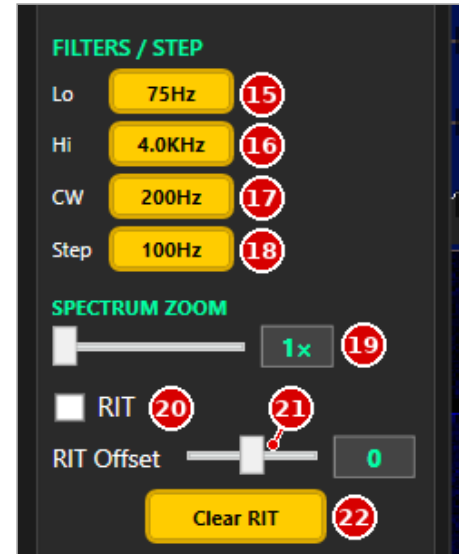
18. **Step**. Tuning step for the mouse wheel on the MAIN tab: 1 Hz → 10 Hz → 100 Hz → 1 kHz → 10 kHz → 100 kHz.

19. **SPECTRUM ZOOM** (1×–4×). Magnifies the spectrum and waterfall around your frequency.

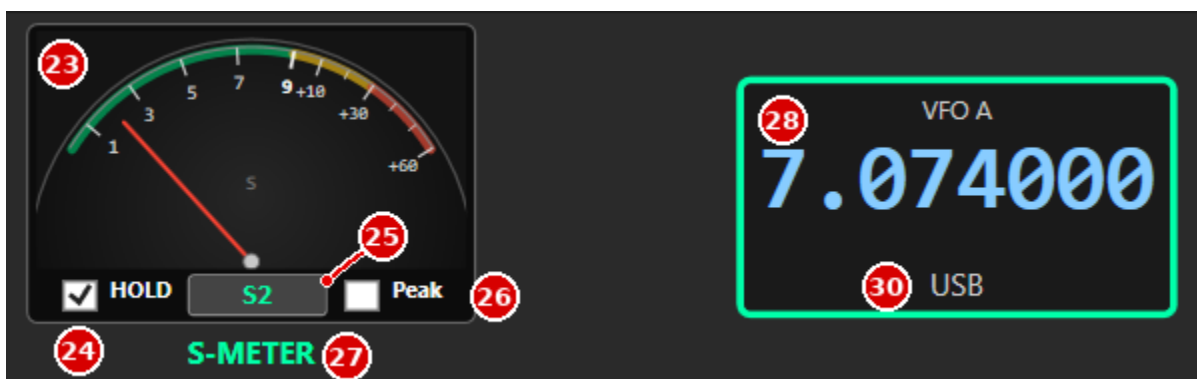
20. **RIT** checkbox. Turns receive offset on for the selected VFO.

21. **RIT Offset** slider (–500 to +500 Hz). The wheel moves it 10 Hz per notch.

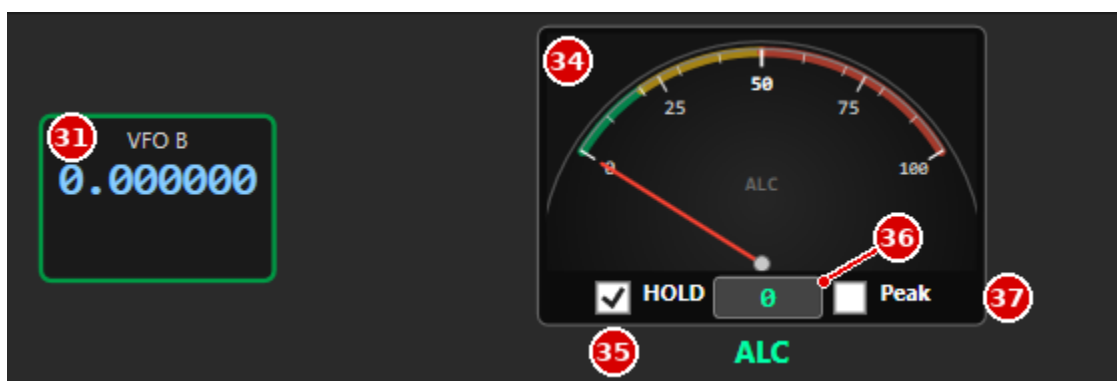
22. **Clear RIT**. Turns RIT off and sets the offset back to 0.



MSCC Operator's Guide — Windows (Preliminary) Meters and VFOs (top of the center column)



S-meter and VFO A, items 23–30



VFO B and ALC meter, items 31–37

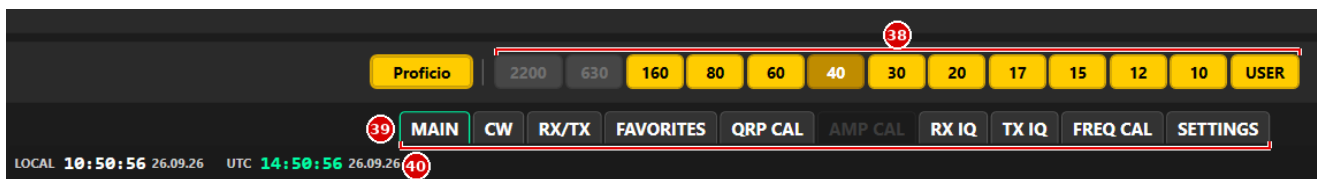
- 23. **S-meter.** Received signal strength, S0 to S9+.
 - If you use the optional external SWR meter (Settings), the meter shows forward power while you transmit, and the reading box shows SWR.
- 24. **HOLD** (S-meter). The needle falls back slowly. On by default.
- 25. **Reading box.** Shows the S reading on receive and SWR on transmit.
 - On an SWR fault it turns into a red **RESET** button. Click it to clear the fault. Transmit is blocked until you do.
- 26. **Peak** (S-meter). Shows an orange peak needle that holds the highest reading for about 2 seconds.
- 27. **Meter label.** Shows **S-METER**, **FWD / SWR** or **SWR FAULT**.
- 28. **VFO A.** Click it to make VFO A active; the active VFO has a bright border. To tune:
 - Put the mouse pointer over a digit and roll the wheel. That digit changes, and the digits to its right go to zero.
 - Over the 1 kHz digit, for example, you tune in 1 kHz steps.
- 29. **VFO A RIT readout.** The RIT offset in Hz (yellow).
- 30. **VFO A mode.** The current mode (LSB, USB, DIG-U, CW, AM, FM, TUNE).



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31. **VFO B.** Works the same way as VFO A. Each VFO remembers its own band. In FM repeater mode VFO B shows the transmit frequency.
32. **VFO B RIT readout.**
33. **VFO B mode.**
34. **ALC meter.** Transmit ALC level. It rests at zero on receive.
35. **HOLD (ALC).** Slow needle fall.
36. **ALC reading box.** The ALC number.
37. **Peak (ALC).** Peak needle held for about 2 seconds.

Band buttons



Band buttons, tab strip and clock, items 38–40

The bands you can use depend on your radio. MSCC works this out from the radio's firmware. Bands your radio can't use are grayed out: Proficio and Ultimus are HF radios (160–10 m), and Geminus is an LF/MF radio (2200 m and 630 m).

38. **Band buttons: 2200, 630, 160, 80, 60, 40, 30, 20, 17, 15, 12, 10 and USER.** Click a band to go there. On your first visit each band starts at:
 - 2200 m: 136.000 kHz; 630 m: 474.200 kHz.
 - 160: 1.800 MHz; 80: 3.500; 60: 5.350; 40: 7.000; 30: 10.100; 20: 14.000; 17: 18.100; 15: 21.000; 12: 24.900; 10: 28.000 MHz.
 - After that, each band button returns to the frequency, mode and filters you last used on that band, separately for VFO A and VFO B.
 - The active band button is darker.
 - With no saved mode, MSCC uses LSB below 10 MHz and USB at 10 MHz and above.
 - **USER** (general coverage and time stations). The first click goes to general coverage. Each further click steps through the presets:
 - HF radios: WWV 5, 10, 15 and 20 MHz; CHU 3.330 and 7.850 MHz; RWM 9.996 MHz; USER (your own last general-coverage frequency).
 - Geminus: 198, 660 and 880 kHz.
 - These stations are good references for FREQ CAL.



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Tabs

39. **Tab strip:** MAIN, CW, RX/TX, FAVORITES, QRP CAL, AMP CAL, RX IQ, TX IQ, FREQ CAL, SETTINGS.

- QRP CAL and TX IQ are grayed while AMP is on. AMP CAL is grayed while AMP is off.
- When you leave a calibration tab, MSCC restores the frequency, mode, filters and power you had before.

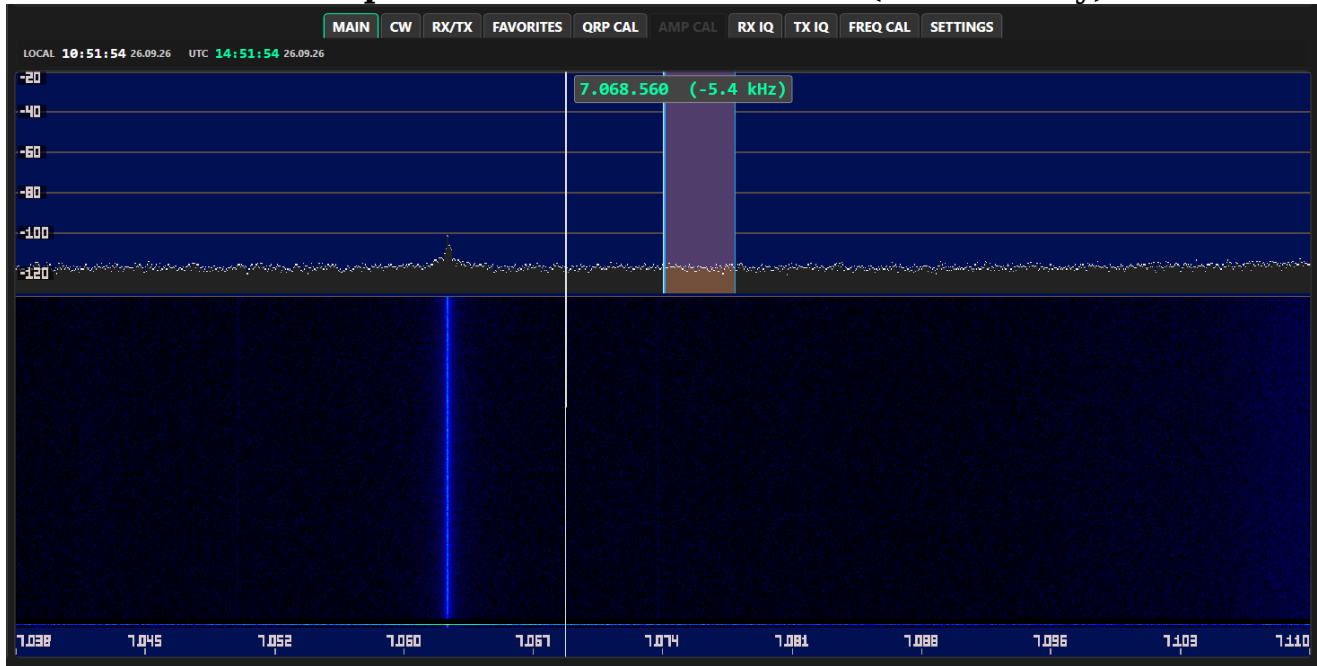
MAIN tab

40. **Clock.** Local and UTC time and date above the spectrum. Turn it on with TIME DISPLAY in the S/W window.

41. **Spectrum and waterfall.**

- **Left-click** a signal to tune to it. With AUTO SNAP on (S/W window), the frequency rounds to 1 kHz, 500 Hz or 100 Hz (not in CW).
- **Right-click** marks the nearest signal peak and shows its frequency and level at the top left. This needs PEAK MARKER on in the S/W window.
- **Hover** to see a cursor line with the frequency under the mouse and its offset from center.
- **Mouse wheel** on the MAIN tab, when the pointer isn't over a control: tunes VFO A by the Step value.
- The shaded band on the spectrum is your receive passband.

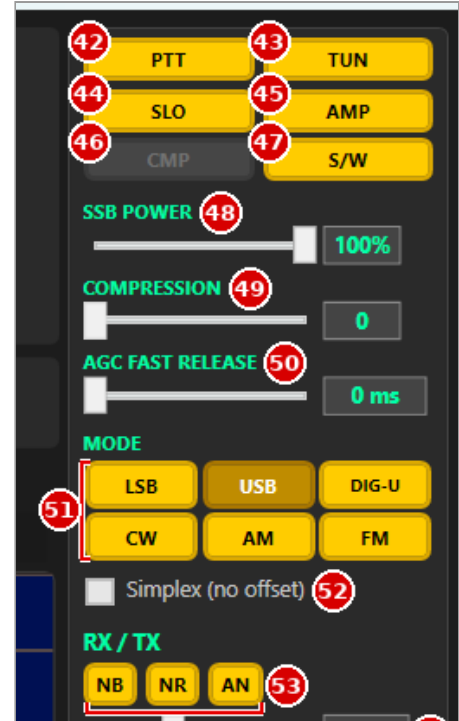
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MAIN tab: hovering over the spectrum shows the cursor line with its frequency and offset from center

Right rail

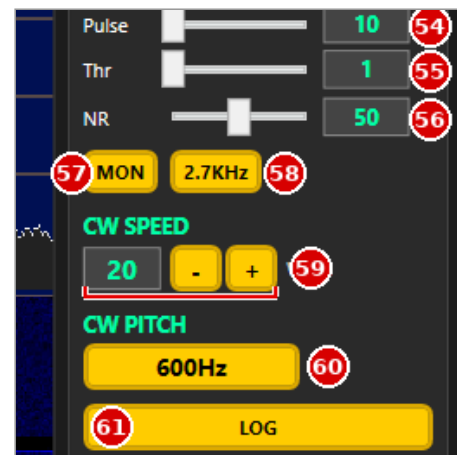
42. **PTT**. Transmit on/off; red while on. It is grayed if the servers have taken over transmit (for example a hardware PTT or a CAT program), and blocked during an SWR fault.
43. **TUN**. Sends a steady carrier at Tune Power for your tuner or amplifier; red while on. Click again to return to your previous mode.
44. **SLO / MED / FST** (AGC). Cycles the AGC speed.
45. **AMP**. Switches between the radio's two power tables so you can go from barefoot (QRP) to an external amplifier (QRO) and back without overdriving the amplifier. Red while on.
 - **AMP off** = normal QRP operation, up to the radio's full 5 W, using the QRP CAL power table.
 - **AMP on** = the radio uses the **AMP CAL** power table. That table is set up once on the AMP CAL tab (2.8) to drive your amplifier correctly.
 - AMP on and AMP off each remember their own Tune Power.
 - AMP also decides which calibration tabs are available: QRP CAL and TX IQ need AMP off, AMP CAL needs AMP on.



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▲ WARNING: Don't overdrive your amplifier. Multus amplifiers are designed to give full output with about 75 % of the radio's power. Feeding the full 5 W QRP output into the amplifier overdrives it. Always turn **AMP on** when the amplifier is in line, and calibrate the AMP table first (2.8).

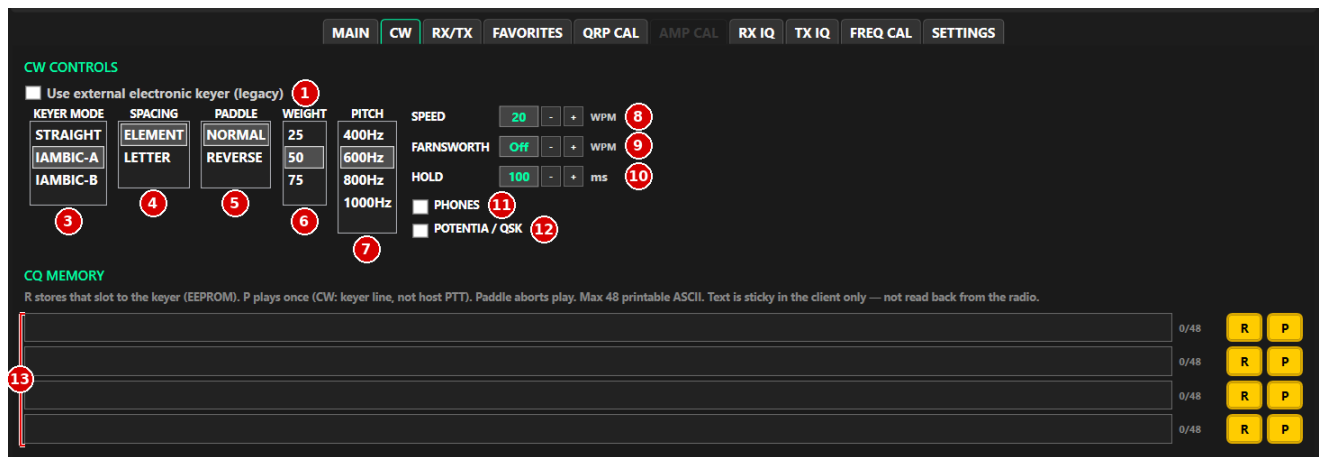
46. **CMP.** Speech compressor on/off. Grayed on the Digital path.
47. **S/W.** Opens the Spectrum / Waterfall Controls window (2.13).
48. **Power slider** (0–100 %). The label changes to match what you're doing: **TUNE POWER** (TUN on), **CW POWER**, **AM CARRIER**, **FM POWER**, or **SSB POWER** (LSB, USB and DIG-U, with any audio setting). **TUNE POWER** is used only for TUNE; no other mode uses it. These are the same settings as on the RX/TX tab. The number is not watts: it is a percentage of the QRP CAL level you set (2.7).
49. **COMPRESSION** (0–24). Compressor amount. Grayed on the Digital path.
50. **AGC FAST RELEASE** (0–2000 ms, default 500). How quickly the AGC recovers.
51. **Mode buttons: LSB, USB, DIG-U, CW, AM, FM.** The active mode is darker. Each mode remembers its own filters.
 - **DIG-U** is upper sideband for digital modes. It switches audio to the Digital path and gives the extra 1.0k/1.4k Hi filters.
 - **FM** is narrow FM. By default it transmits 100 kHz below the receive frequency (repeater split), shown on VFO B.
52. **Simplex (no offset).** FM only. Check it to transmit on the receive frequency.
53. **NB / NR / AN.** Noise blanker, noise reduction and automatic notch, each on/off.
54. **Pulse** (10–510). Noise blanker pulse width.
55. **Thr** (1–1009). Noise blanker threshold. Question for Ron: units
56. **NR level** (0–100, default 50). Noise reduction strength.
57. **MON.** Lets you hear your own transmit audio.
58. **TX BW.** Transmit bandwidth: 2.4 → 2.7 → 3.0 → 5.5 kHz.



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59. **CW SPEED** – / +. Keyer speed in WPM. – slows down (to 5 WPM), + speeds up (to 60 WPM).
60. **CW PITCH**. 400 → 600 → 800 → 1000 Hz. The CW tone you hear, and where CW signals sit in the filter.
61. **LOG**. Opens the Debug Log window (2.15).

2.4 CW Tab



CW tab

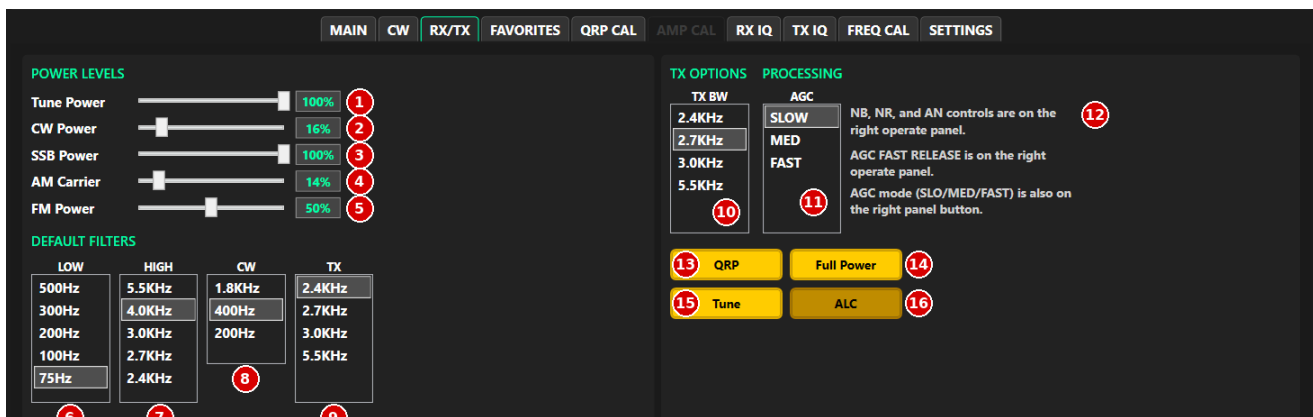
Items 2 and 14 have no callout; they are shown only when needed (2 when box 1 is checked, 14 while a memory action is running).

1. **Use external electronic keyer (legacy)**. Check this if you key the radio with your own external keyer, or you have a Legacy (non-MKII) radio.
 - The radio's built-in keyer controls gray out. HOLD still works.
 - The servers read this setting only when they start. MSCC offers to Stop and Start for you. For a remote radio it explains what to change on the radio computer.
2. **Legacy note** (orange). Shown when box 1 is checked.
3. **KEYER MODE**. STRAIGHT, IAMBIC-A (default) or IAMBIC-B.
4. **SPACING**. ELEMENT or LETTER.
5. **PADDLE**. NORMAL or REVERSE (swaps dot and dash).
6. **WEIGHT**. 25, 50 (default) or 75.
7. **PITCH**. 400, 600, 800 or 1000 Hz. Same as CW PITCH on the right rail.
8. **SPEED** – / +. 5–60 WPM. Same as the right rail.
9. **FARNSWORTH** – / +. Off, or 5–60 WPM. It adds extra space between letters and words **when playing CQ memories only**; your paddle sending isn't affected.
10. **HOLD** – / +. 1–500 ms in steps of 10 (default 100). How long the radio stays in transmit after you stop keying.

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11. **PHONES.** **Question for Ron:** what PHONES does today
12. **POTENTIA / QSK.** **Question for Ron:** what POTENTIA / QSK does today
13. **CQ memory slots 0–3.** Each slot has a text box, a character count, **R** and **P**.
 - Type up to 48 characters per slot.
 - **R** stores the text in the radio's keyer.
 - **P** plays it once. Touch the paddle to stop playback.
 - R and P work only while connected. See Appendix B.
14. **Memory status line.** For example “Storing slot...” or “Not connected — Start session first”.

2.5 RX/TX Tab



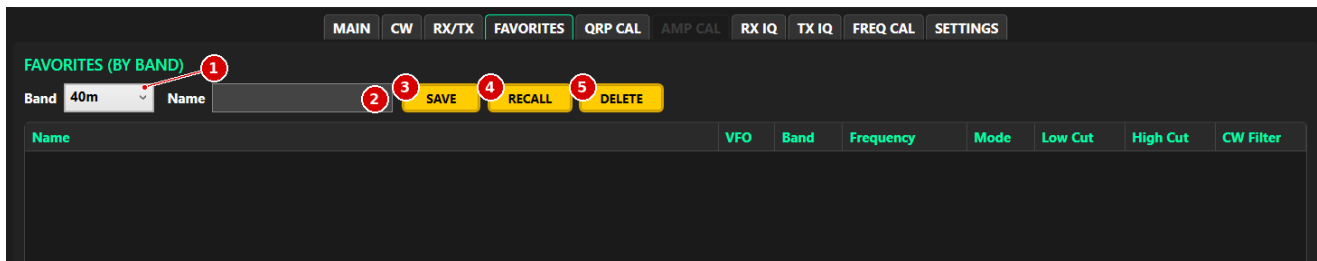
RX/TX tab

1. **Tune Power** (0–100 %). Sets the drive for TUNE only (the TUN carrier). No other mode uses it. It is kept separately for AMP on and AMP off.
2. **CW Power** (0–100 %).
3. **SSB Power** (0–100 %). LSB, USB and DIG-U, with any audio setting (Phones, Digital or Remote).
4. **AM Carrier** (0–100 %).
5. **FM Power** (0–100 %).
6. **DEFAULT FILTERS** → **LOW**. 500 / 300 / 200 / 100 / 75 Hz.
7. **DEFAULT FILTERS** → **HIGH**. 5.5k / 4.0k / 3.0k / 2.7k / 2.4k.
8. **DEFAULT FILTERS** → **CW**. 1.8k / 400 / 200 Hz.
9. **DEFAULT FILTERS** → **TX**. 2.4k / 2.7k / 3.0k / 5.5k.
 - Items 6–9 are the radio's stored default filters. They are applied each time MSCC starts. The filters you use day to day are the Lo/Hi/CW buttons on the left rail.
10. **TX BW**. Same as the TX BW button on the right rail.
11. **AGC**. SLOW / MED / FAST. Same as the AGC button on the right rail.
12. **Notes**. Reminder that NB, NR and AN are on the right rail.

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13. **QRP.** Barefoot power, the same as **AMP** off (2.3 item 45). The output is your power setting as a percentage of your QRP CAL level, so 100 % gives the 5 W you set in QRP CAL (2.7).
14. **Full Power.** The amplifier (QRO) setting, the same as **AMP** on (2.3 item 45). The radio uses the AMP CAL table for the band (2.8), which lowers the drive for the amplifier; it never drives the radio above your QRP CAL level. If the band has no AMP CAL yet, the amplifier gets the full QRP drive, so do AMP CAL first (see the warning in 2.8).
15. **Tune.** Same as TUN on the right rail.
16. **ALC.** Transmit automatic level control on/off. Keep ALC on for digital modes, and set the digital level so the ALC meter is just into the yellow.

2.6 FAVORITES Tab



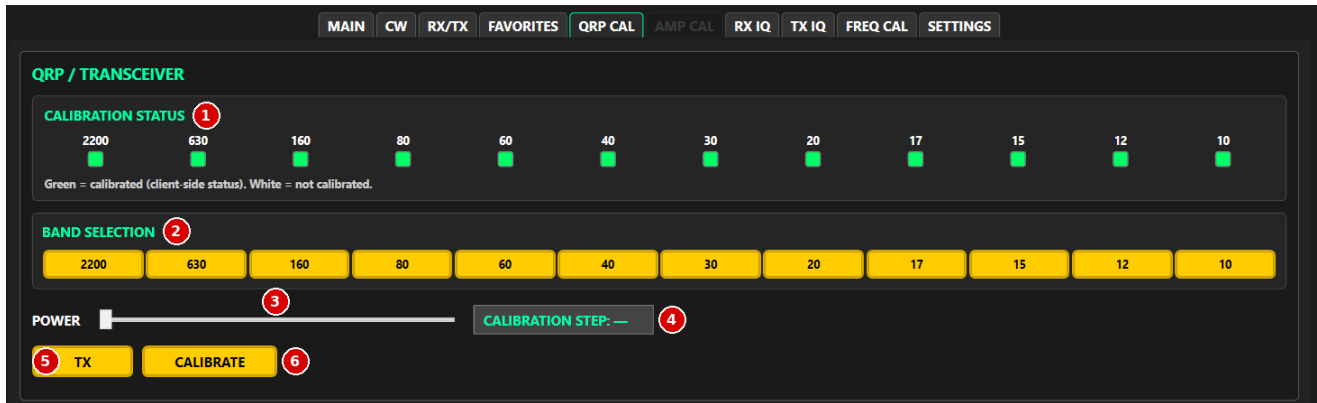
FAVORITES tab

1. **Band.** Shows the favorites saved for one band.
2. **Name.** Name for a new favorite (up to 32 characters).
3. **SAVE.** Saves the current frequency, mode, filters and VFO under this name, on the band you are on now. If the name already exists, MSCC asks before replacing it.
4. **RECALL.** Returns to the selected favorite.
5. **DELETE.** Removes the selected favorite, after asking.
6. **Favorites list.** Name, VFO, band, frequency, mode and filters. Click a row to select it.

Favorites are kept on this computer only (file MSCC_Favorites.ini).

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2.7 QRP CAL Tab (power calibration; AMP must be off)

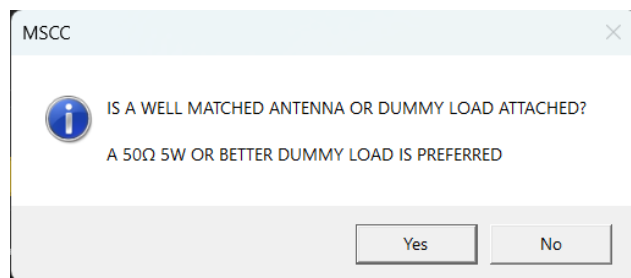


QRP CAL tab

This tab sets the radio's output power on each band.

- You need a 50 Ω dummy load rated for 5 W or more, and a wattmeter.
- **Do not set the power above 5 W. More than 5 W causes spurious emissions.** This applies to all current models.

1. **Calibration status lamps.** One per band. Green = calibrated, white = not yet.
2. **Band buttons.** Pick the band to calibrate; the radio tunes to that band's calibration frequency.
3. **POWER slider** (0–100). Adjusts the output. Works only while CALIBRATE is on.
4. **CALIBRATION STEP.** The current setting as reported by the radio.
5. **TX.** Sends a test carrier; red TX ON while on. The first time, MSCC asks you to confirm a dummy load or matched antenna is connected.
6. **CALIBRATE.** Starts calibration: the radio transmits and the POWER slider comes alive.



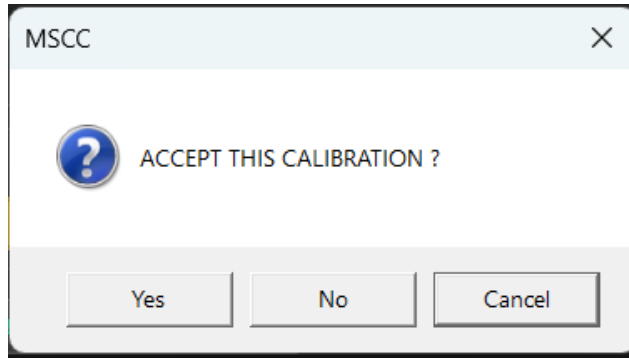
The first time you transmit here, MSCC asks you to confirm that a dummy load or matched antenna is connected

Procedure:

1. Turn AMP off.
2. Connect the dummy load and wattmeter.

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3. Open QRP CAL and pick a band (2).
4. Click **CALIBRATE** (6).
5. Move **POWER** (3) until the wattmeter reads 5 W (not more).
6. Click **CALIBRATE** again. MISC asks “ACCEPT THIS CALIBRATION?”:

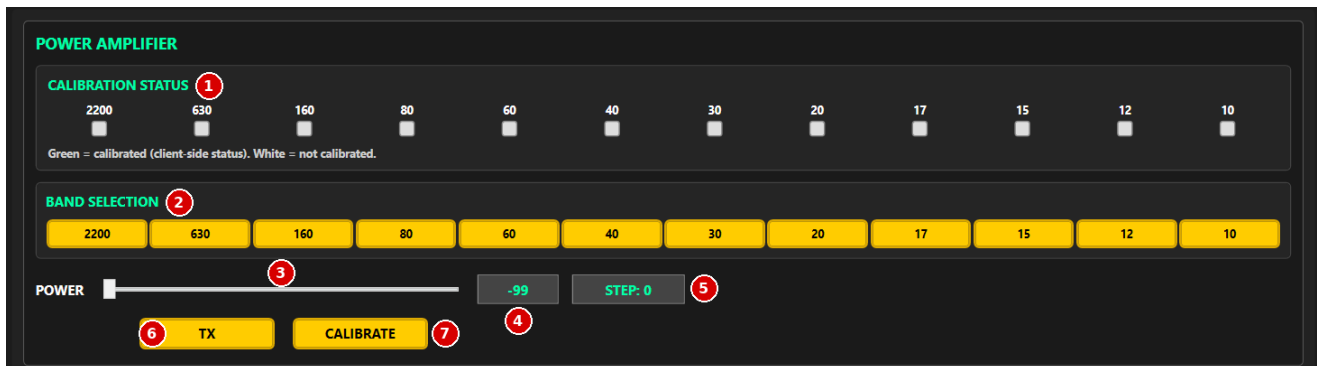


Accept this calibration?

- **Yes** saves it and the lamp turns green.
- **No** leaves the band uncalibrated.
- **Cancel** puts back the previous setting.

7. Repeat for each band.

2.8 AMP CAL Tab (AMP must be on)



AMP CAL tab

Use this tab to set up the **AMP power table**, the drive level the radio uses when **AMP** is on. Once each band is calibrated here, you can switch between QRP (AMP off, 5 W) and QRO (AMP on, amplifier) without overdriving the amplifier. A band must pass QRP CAL (green lamp) before you can AMP CAL it.

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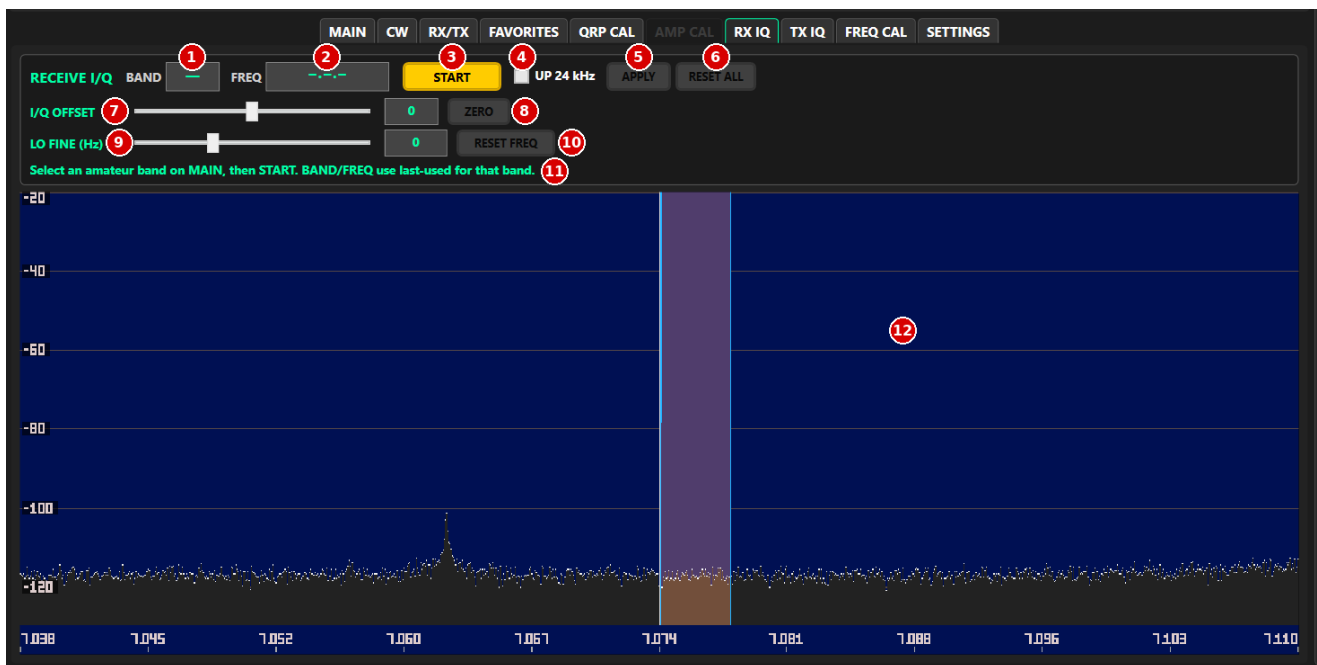
⚠ WARNING: Multus amplifiers reach full output with about 75 % of the radio's power. Full 5 W QRP drive overdrives them. Calibrate AMP CAL for each band you use with the amplifier, and keep **AMP on** whenever the amplifier is in line.

1. **Status lamps.** Green = calibrated.
2. **Band buttons.**
3. **POWER slider** (-99 to 0). Works only while CALIBRATE is on.
4. **Value readout.**
5. **STEP readout** (1–100).
6. **TX.** Full tune carrier through the amplifier path.
7. **CALIBRATE.** Same Yes / No / Cancel choice as QRP CAL.

Popup pictures coming: "QRP not calibrated for this band" and "Accept this calibration?"

Target: with CALIBRATE on, adjust **POWER** until the amplifier just reaches its rated output.

2.9 RX IQ Tab (receive image balance)



RX IQ tab

First choose an amateur band on the MAIN tab. This doesn't work in general coverage (GEN).

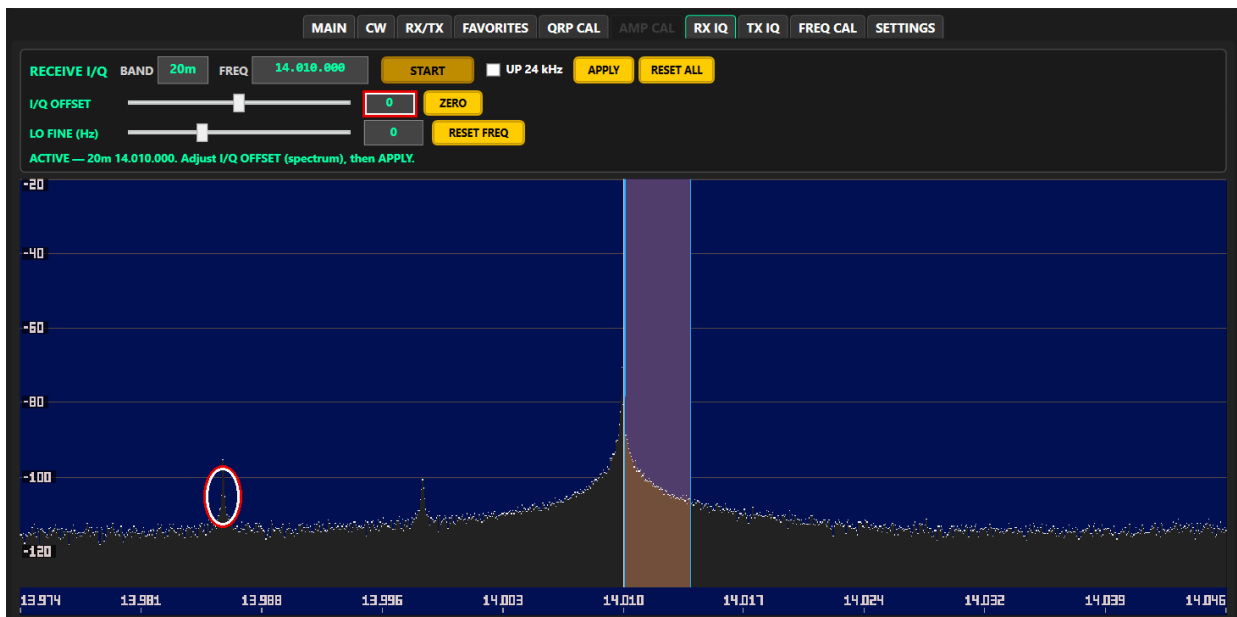
1. **BAND.** Band being balanced.
2. **FREQ.** Frequency in use.

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3. **START / ACTIVE.** Starts or ends the session. The other controls work only while ACTIVE.
4. **UP 24 kHz.** Shifts the receiver up 24 kHz to help you find the image.
5. **APPLY.** Saves the value for this band, after asking.
6. **RESET ALL.** Copies the current value to **all** bands, after a warning.
7. **I/Q OFFSET** (–200 to +200). Adjust for the smallest image on the spectrum.
8. **ZERO.** Sets the offset to 0 and saves it.
9. **LO FINE** (–1000 to +2009 Hz). Fine retune during the session.
10. **RESET FREQ.** Clears LO FINE.
11. **Status line.**
12. **Spectrum.** Live view to watch the image. Clicking here doesn't tune.

Example: before, after, and saving. These captures were made on 20 m at 14.010.000. The red ring marks where the image signal shows up. The red box marks the I/Q OFFSET value.

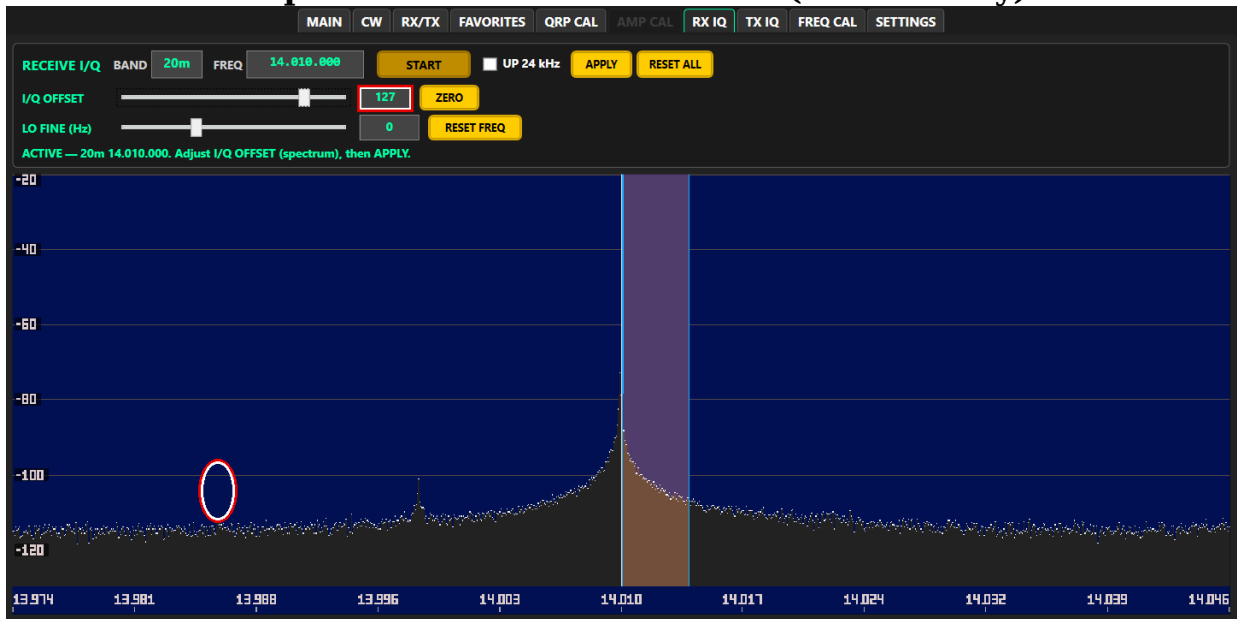
1. **Before.** Press START, then look for the image. In this example, with I/Q OFFSET at 0, a small unwanted image signal shows up near 13.986.



RX IQ before adjustment: I/Q OFFSET 0, image signal visible

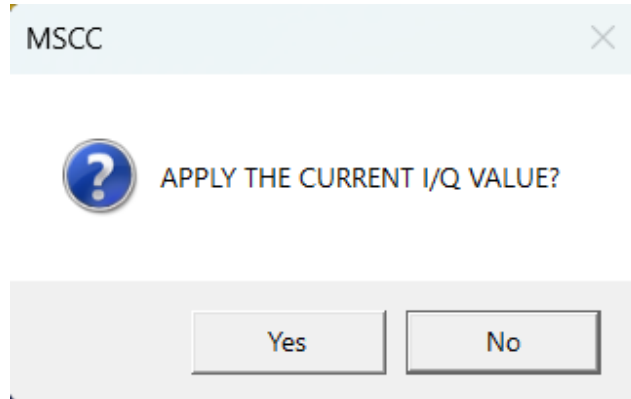
2. **After.** Move the I/Q OFFSET slider until the image drops into the noise. Here that happened at 127. The best value is different for every radio and band.

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RX IQ after adjustment: I/Q OFFSET 127, image signal gone

3. **Save.** Press APPLY. When MSCC asks “**APPLY THE CURRENT I/Q VALUE?**”, choose **Yes** to save the value for this band or **No** to keep the old one.

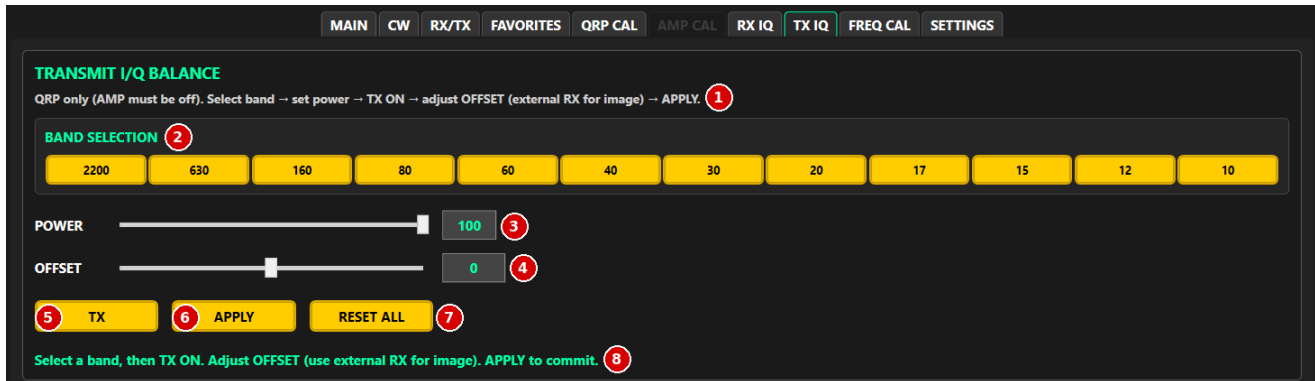


Apply I/Q value confirmation

Popup pictures coming: Reset all bands? (RESET ALL warning)

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2.10 TX IQ Tab (transmit image balance; AMP must be off)



TX IQ tab

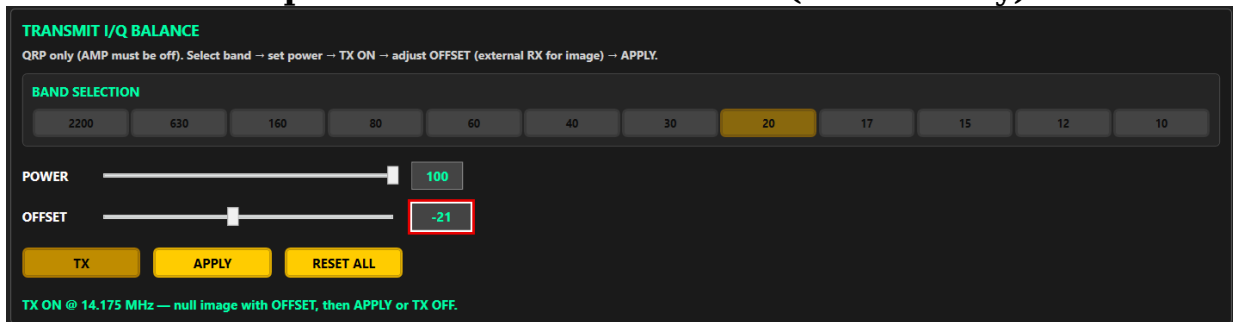
You need a second receiver to hear or see the transmit image. Use a dummy load, and keep the second receiver's antenna well away so it doesn't overload. Tune the second receiver about 24 kHz from the displayed frequency to find the image.

1. **Instructions.** Select band → set power → TX ON → adjust OFFSET → APPLY.
2. **Band buttons.** Tune to that band's TX IQ frequency. Locked while TX is on.
3. **POWER** (0–100, default 100).
4. **OFFSET** (–200 to +200). Adjust for the weakest image. Works only while TX is on.
5. **TX / TX ON.** Starts or stops the test carrier.
6. **APPLY.** Saves the value for this band, after asking.
7. **RESET ALL.** Returns **all** bands to the factory values, after a warning. TX must be off.
8. **Status line.**

Example: before, after, saving, and reset. These captures were made on 20 m (test carrier at 14.175 MHz, POWER 100). The red box marks the OFFSET value. The image shows up on your second receiver, not in MSCC.

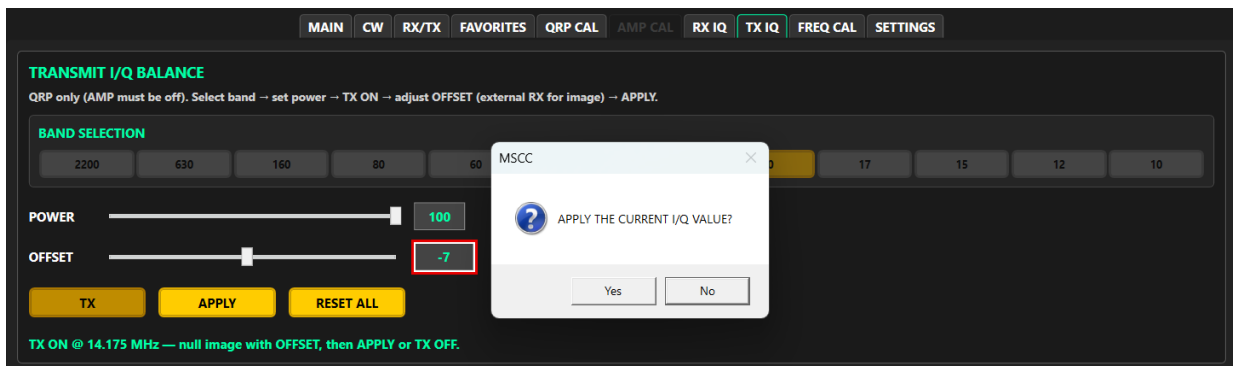
1. **Before.** Select the band and press TX. The status line reads "TX ON @ 14.175 MHz". In this example OFFSET started at –21.

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TX IQ before adjustment: TX on, OFFSET -21

2. **After and save.** Watch the second receiver and move OFFSET until the image is weakest. Here that was at -7. The best value is different for every radio and band. Press APPLY. When MSCC asks “**APPLY THE CURRENT I/Q VALUE?**”, choose **Yes** to save the value for this band or **No** to keep the old one. Press TX again to turn the carrier off.

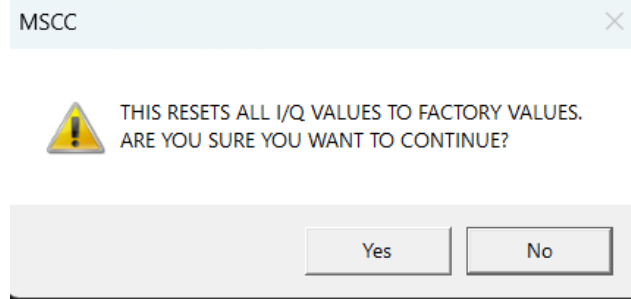


TX IQ after adjustment: OFFSET -7, with the Apply confirmation

3. **Reset (only if needed).** With TX off, RESET ALL shows this warning: “**THIS RESETS ALL I/Q VALUES TO FACTORY VALUES. ARE YOU SURE YOU WANT TO CONTINUE?**” Choose **Yes** only if you want to throw away the saved values for **every** band. Choose **No** to cancel.



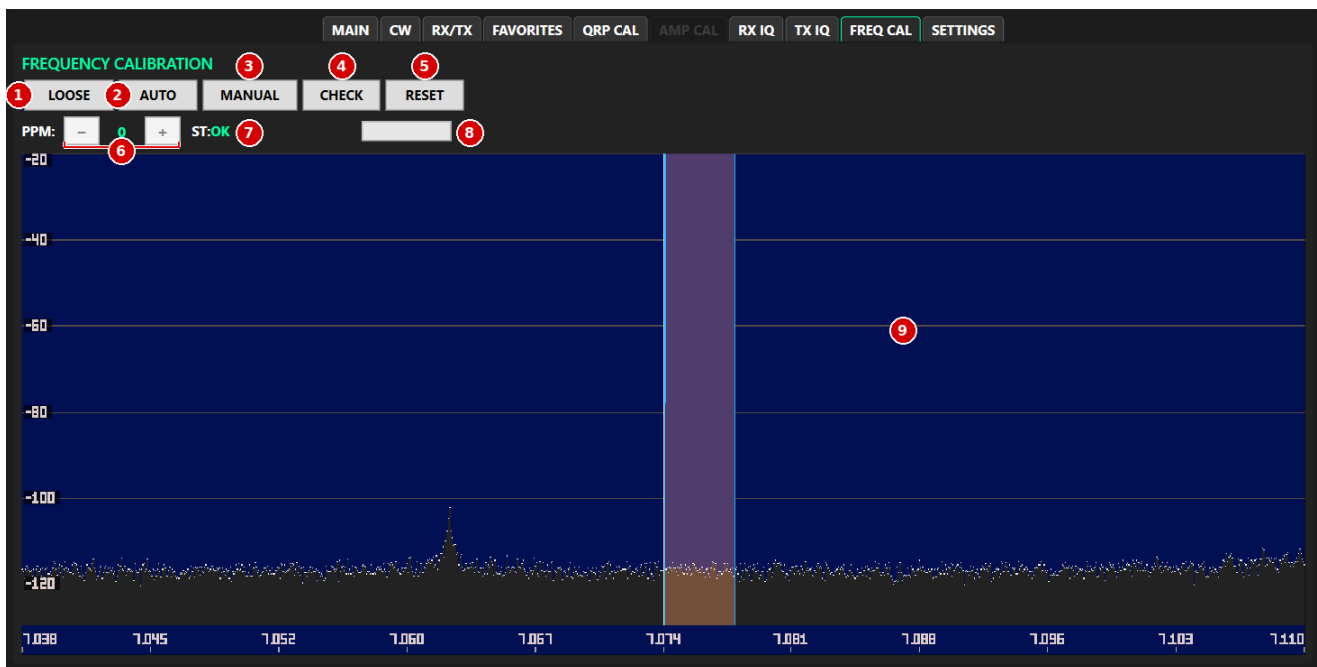
MSCC Operator's Guide — Windows (Preliminary)



Reset all TX I/Q values warning

2.11 FREQ CAL Tab (frequency calibration)

While a calibration is running, don't change tabs or settings until it finishes. The tab shows a reminder during the run. MSCC won't let you open QRP CAL, AMP CAL or TX IQ until the run is done.



FREQ CAL tab

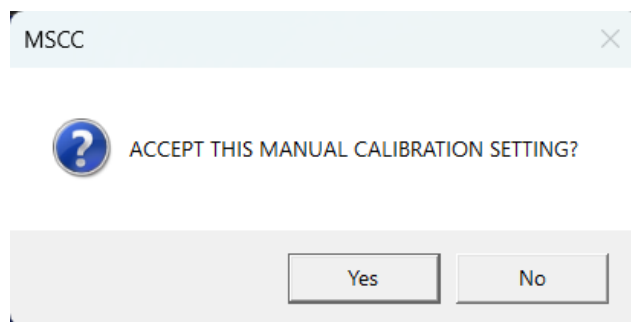
Mode on this tab. Opening FREQ CAL switches the radio to **CW** with a **600 Hz pitch** and the **200 Hz CW filter**, which is what the measurement needs. The mode button shows CW. When you leave the tab, MSCC puts back your mode, filter and pitch (DIG-U comes back as DIG-U). If you go back to MAIN while a run is going, your mode comes back when the run finishes. While MANUAL is on, you can't leave the tab; finish MANUAL first.

1. **LOOSE / TIGHT.** How close the measurement must get before it calls the job done. AUTO and CHECK both use this setting.

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2. **AUTO.** Automatic calibration against a steady carrier on your current frequency. MSCC asks:
 - **Yes** = COARSE (± 250 Hz in 5 Hz steps).
 - **No** = FINE (± 50 Hz in 1 Hz steps).
 - **Cancel** = don't start.
3. **MANUAL.** Manual adjustment with PPM - / +. The tab is already in CW with the 600 Hz pitch, so the carrier is heard as a 600 Hz tone. Click MANUAL again to finish; MSCC asks whether to keep the setting.
4. **CHECK.** Measures the error without changing anything. CHECK only looks within ± 50 Hz of the dial. If the radio is further off than that (for example right after RESET), CHECK reports FAILED and suggests running AUTO (COARSE) first. A failed CHECK leaves your mode unchanged.
5. **RESET.** Returns to the factory calibration, after asking.
6. **PPM - / value / +.** Manual adjustment, in MANUAL mode only (range ± 100 PPM).
7. **Status.** RUNNING (AUTO) or CHECKING (CHECK) in amber, then COMPLETED (green) with a number in Hz, or FAILED (red). After CHECK, the number is the radio's error right now. After AUTO, it's the error AUTO found **before** it corrected it (see the example below). While a run is going, a reminder line under the status reads "Calibration running — don't change tabs or settings until it finishes."
8. **Progress bar.** Starts empty at the beginning of each run and fills as the run goes.
9. **Spectrum.** Live view. Clicking here doesn't tune.

When you click MANUAL again to finish, MSCC asks "**ACCEPT THIS MANUAL CALIBRATION SETTING?**" Choose **Yes** to keep the new setting or **No** to discard it.



Accept manual calibration setting confirmation

Popup pictures coming: reset to factory?

Procedure (adapted from the 2022 guide):

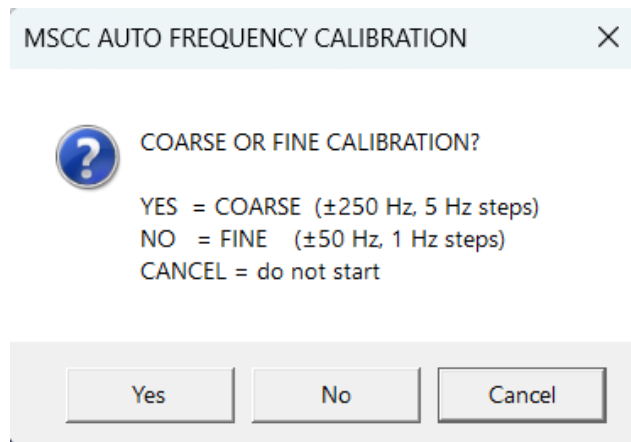
1. On the band bar, click **USER/GEN** until you reach a strong time station, S7 or better. WWV or CHU on HF; 198/660/880 kHz on Geminus.

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2. Open **FREQ CAL**. The tab switches to **CW** (600 Hz pitch, 200 Hz filter) for you; you don't need to change the mode.
3. Click **AUTO** and choose **Yes (COARSE)**. Wait for **AUTO COMPLETED**.
4. Click **AUTO** again and choose **No (FINE)**. Wait for **AUTO COMPLETED**.
5. Click **CHECK** to confirm. After **FINE**, **CHECK COMPLETED** with an error of a few Hz (for example -1 Hz) is a good result.
6. Stay on this tab until each run finishes. When you leave, your mode, filter and pitch come back. A **FAILED** run leaves your mode unchanged.

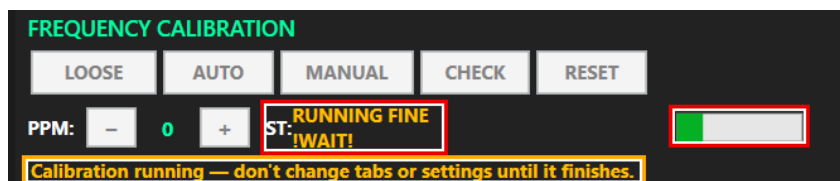
Example: AUTO COARSE, AUTO FINE, then CHECK. These captures were made on 20 m at 14.010.000. Red boxes mark the status text and the progress bar. The amber box marks the reminder that shows while a run is going.

1. **AUTO, COARSE first.** Click **AUTO**. MSCC asks which kind of run you want. Choose **Yes** for **COARSE**. While **COARSE** runs, the tab looks like the **FINE** run in step 2. Wait for **AUTO COMPLETED**.



AUTO asks COARSE (Yes), FINE (No) or Cancel

2. **AUTO FINE.** Click **AUTO** again and choose **No (FINE)**. The status reads **RUNNING FINE / !WAIT!** in amber, the progress bar fills from empty, and the reminder line appears.



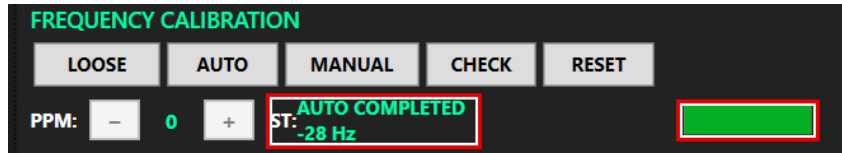
AUTO FINE running: amber status, progress bar filling, reminder line

When the run ends, the status turns green with **AUTO COMPLETED**, and the bar is full. The number (here -28 Hz) is the error **AUTO found before** it corrected the radio, not the error left afterwards. A **CHECK** just before this run also measured -28 Hz, and **AUTO**



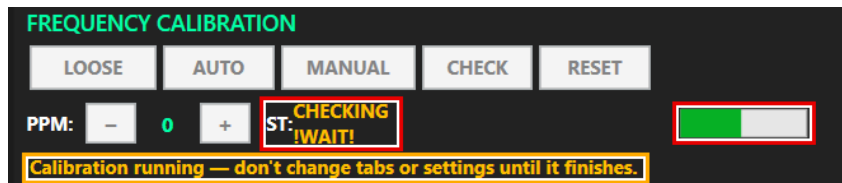
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fixed that whole error. **Known display issue:** the screen doesn't say "corrected", so -28 Hz looks like a new error. A clearer label is planned. Run CHECK to see where you are now.



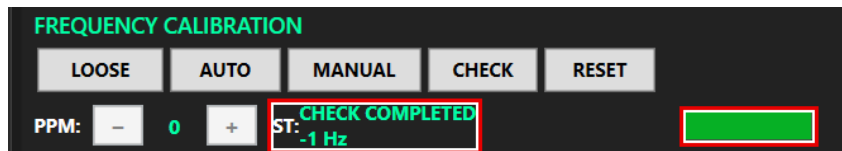
AUTO FINE finished: AUTO COMPLETED, -28 Hz, bar full

3. **CHECK.** Click **CHECK**. The status reads **CHECKING / !WAIT!** and the bar starts again from empty.



CHECK running: amber status, progress bar filling, reminder line

CHECK passed, within 1 Hz. CHECK COMPLETED in green shows the radio's error now: here -1 Hz after the FINE run, so the calibration is good. A failure shows **CHECK FAILED** in red.



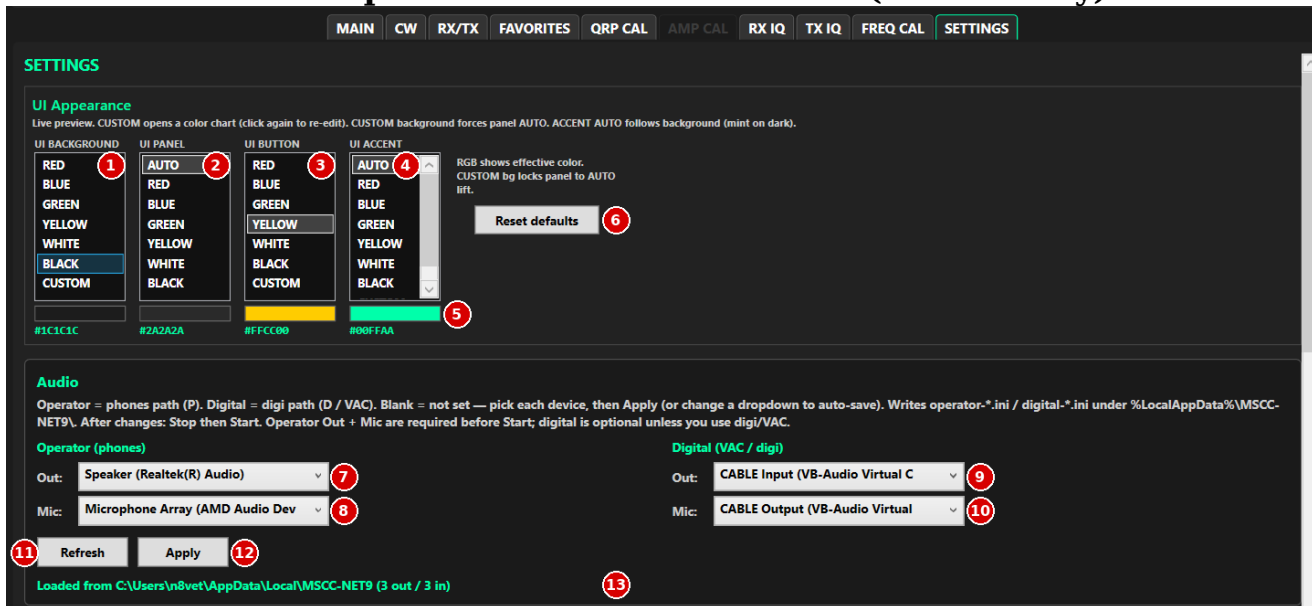
CHECK passed after FINE: CHECK COMPLETED, -1 Hz

All pictures come from one session. The "CHECK running" picture is from the CHECK just before the FINE run. The -1 Hz result is the CHECK after it.

2.12 SETTINGS Tab

The SETTINGS page scrolls, so it is shown in three pictures.

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SETTINGS tab, part 1: UI Appearance and Audio

UI Appearance (changes show immediately)

1. **UI BACKGROUND.** Window background color, or CUSTOM.
2. **UI PANEL.** Panel color; AUTO is a little lighter than the background.
3. **UI BUTTON.** Button color (YELLOW is the classic Multus gold).
4. **UI ACCENT.** Color of headings, readouts and the VFO border.
5. **Color swatches and codes.** The color actually in use under each list.
6. **Reset defaults.**
 - Choosing CUSTOM opens the Choose Color window (2.16).

Audio (Stop and Start after changes)

7. **Operator (phones) Out.** Your speaker or headphones. Required.
8. **Operator (phones) Mic.** Your microphone. Required.
9. **Digital Out.** For example CABLE Input (VB-Audio). Needed for digital modes.
10. **Digital Mic.** For example CABLE Output.
11. **Refresh.** Looks for audio devices again.
12. **Apply.** Saves your choices (changing a list also saves).
13. **Status.**



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COM Port
Serial port used by ms-sdr (CAT / keyer hardware). Writes %LocalAppData%\MSCC-NET9\comm-port.ini. After changing port: press Stop then Start (green button) so ms-sdr reloads the COM port.

Port: 14 COM5 15 Refresh 16 Apply

Baud: 17 9600 18 PTT via serial pin (PIN=1)

Loaded C:\Users\n8vet\AppData\Local\MSCC-NET9\comm-port.ini 19

SWR Meter
External WiFi SWR (HF/LF). Meter must unicast UDP to this PC. Profile follows Proficio/Geminus. Only one meter powered at a time — one listen port is enough. Auto IP from first packet.

☒ Enable external SWR meter 20

UDP listen port: 21 6999 22 Apply listen

HF (Proficio) **LF (Geminus)**

IP: 192.168.1.124 23 IP: 24

Listening UDP port 6999 25

Radio temperatures
Proficio board temp was used for classic XO drift compensation. MKII TCXO does not need continuous display; kept here for legacy radios and diagnostics. PA temp/current reserved for a future amp-centric UI.

Proficio: 38.0 °C 26

PA temp / PA current — reserved (not shown) 27

SETTINGS tab, part 2: COM Port, SWR Meter and Radio temperatures

COM Port (Stop and Start after changes)

14. **Port.** Serial port the servers use for CAT.
15. **Refresh.**
16. **Apply.**
17. **Baud** (default 9600).
18. **PTT via serial pin (PIN=1).**
19. **Status.**

SWR Meter (optional external WiFi SWR meter)

20. **Enable external SWR meter.**
21. **UDP listen port.**
22. **Apply listen.**
23. **HF (Proficio) IP.** Fills in by itself.
24. **LF (Geminus) IP.** Fills in by itself.
25. **Status.**

Radio temperatures

26. **Transceiver temperature** (°C). Labeled with the radio name, for example “Proficio: 38.0 °C”.
27. **PA temp / current.** Reserved for later.



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Configuration

Reset restores UI/audio/COM templates under %LocalAppData%\MSCC-NET9\ (keeps logs). Radio IQ, freq cal, and QRP are not copied from generic init-files — they reseed from factory tables for the radio connected when servers start. Requires MSCC restart.

Reset configuration to defaults...

28

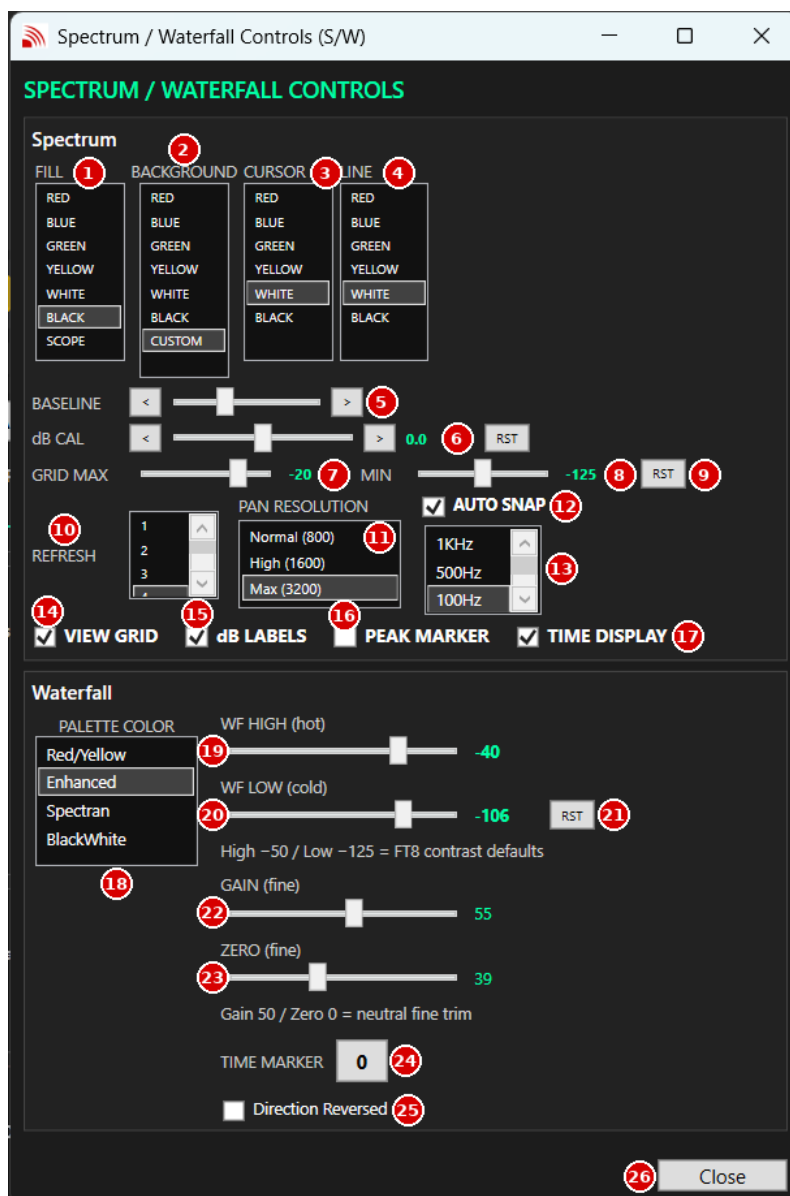
SETTINGS tab, part 3: Configuration

Configuration

28. **Reset configuration to defaults...** Wipes MSCC's settings on this PC (logs are kept) and closes MSCC. Your radio calibration reloads from the factory tables the next time the servers start. Use this as a last resort.

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2.13 Spectrum / Waterfall Controls window (S/W)



Spectrum / Waterfall Controls window

These settings are kept separately for HF and LF bands, and follow the band you are on.

1. **FILL.** Color under the spectrum trace (SCOPE = classic look).
2. **BACKGROUND.** Spectrum background (CUSTOM opens Choose Color).
3. **CURSOR.** Color of the mouse cursor line.
4. **LINE.** Trace color.
5. **BASELINE** < slider >. Raises or lowers the noise floor on the display (50 = middle).
6. **dB CAL** < slider > **readout RST.** Trims the dB scale (± 20 dB).

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7. **GRID MAX** (-80 to 0 dBm, default -20). Top of the display.
8. **MIN** (-160 to -90 dBm, default -125). Bottom of the display.
9. **RST**. Returns the grid to -20 / -125.
10. **REFRESH**. 1, 2, 3, 4, 5, 8, 10 (default 4). How often the spectrum and waterfall are redrawn, and so how fast the waterfall falls. MSCC draws one of every N updates from the radio: 1 = fastest, 10 = slowest. Higher numbers use less computer power.
11. **PAN RESOLUTION**. Normal / High / Max. Sharper display, but more network traffic. Use Normal if a remote display stutters.
12. **AUTO SNAP**. Rounds click-to-tune (not in CW).
13. **Snap step**. 1 kHz / 500 Hz / 100 Hz.
14. **VIEW GRID**.
15. **dB LABELS**.
16. **PEAK MARKER**. Turns on right-click peak marking.
17. **TIME DISPLAY**. Clock above the MAIN spectrum.
18. **PALETTE COLOR**. Waterfall colors: Red/Yellow, Enhanced, Spectran, BlackWhite.
19. **WF HIGH** (-100 to -20, default -50). Signals this strong show in the hottest color.
20. **WF LOW** (-160 to -90, default -125). Background level of the waterfall.
21. **RST**. Returns the waterfall to -50 / -125.
22. **GAIN** (50 = neutral). Fine brightness.
23. **ZERO** (0 = neutral). Fine lift.
24. **TIME MARKER**. 0–5. 0 = off. At 1–5, MSCC draws a white line across the waterfall every 1–5 seconds, labeled with the UTC time (mm:ss), so you can tell when a signal was heard. How fast the waterfall scrolls is set by REFRESH (item 10).
25. **Direction Reversed**. Unchecked (normal) = newest line at the top, right under the spectrum. Checked = newest line at the bottom.
26. **Close**.

2.14 Remote Audio window

Screenshot coming: Remote Audio window (Phones and Digital), callouts 1–15.

This window opens when you turn on **Remote** (left rail). Setting up remote operation is covered in the separate Remote guide.

1. **Phones / Digital**. Choose the remote audio path.
 - Phones = voice with a headset on this computer.
 - Digital = WSJT-X or fldigi on this computer.
2. **Heading and hints**. In Digital, shows the WSJT-X setup and whether CAT is open.

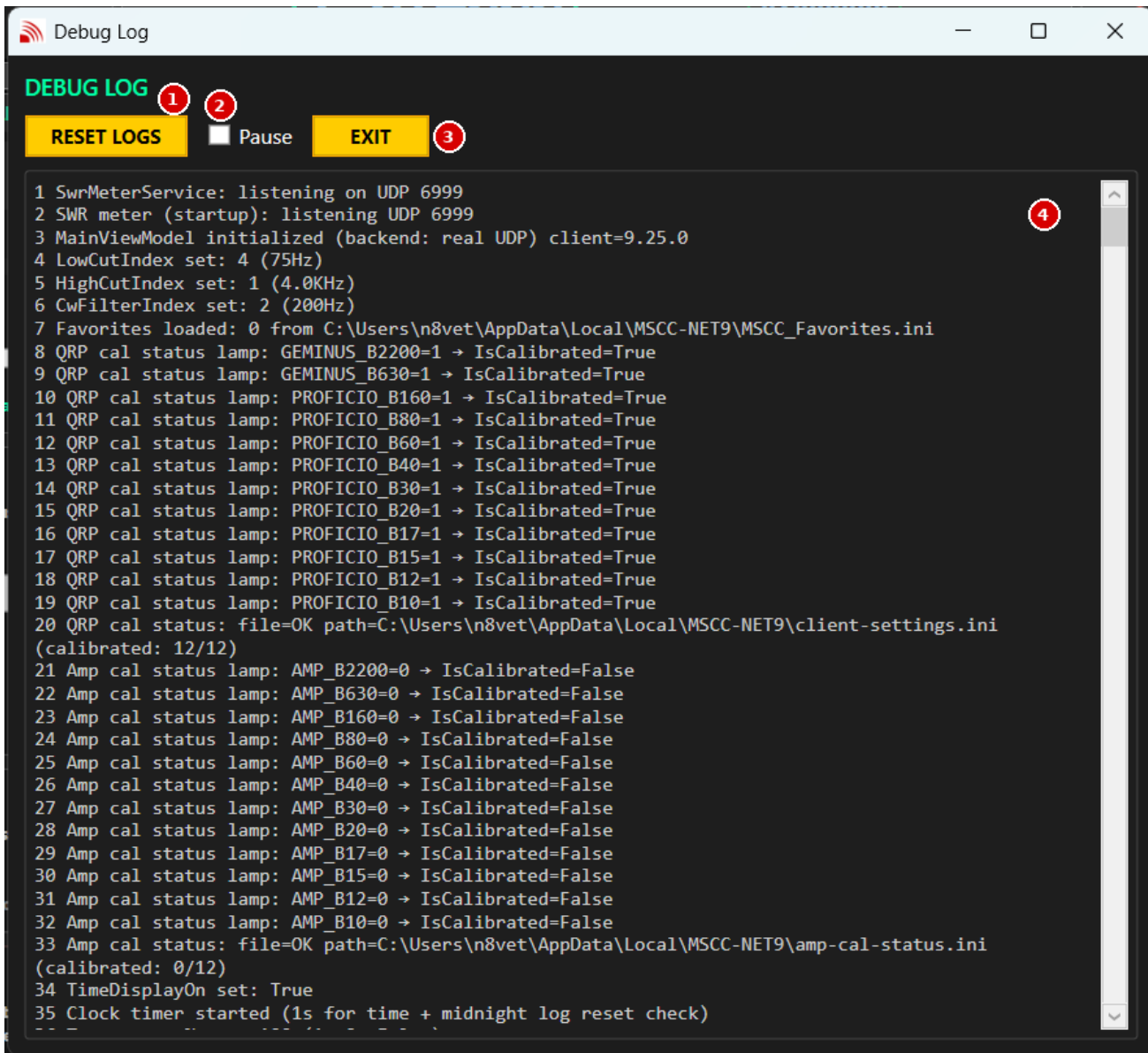


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3. **Device.** Where the radio's audio plays on this computer.
4. **Volume** (0–100, default 80).
5. **Mute phones / Mute VAC play.** Silences playback here.
6. **Monitor at radio.** Keeps the speaker at the radio on as well.
7. **Playback EQ.** Phones only.
8. **Low.** –12 to +12 dB.
9. **Mid.** –12 to +12 dB.
10. **High.** –12 to +12 dB.
11. **Mic.** Microphone (Phones) or virtual cable (Digital) on this computer.
12. **Mic vol** (0–100). Transmit level sent to the radio. It is remembered separately for Phones and Digital. For digital, turn it down if ALC is too high.
13. **Log.** Connection messages.
14. **Reset EQ.**
15. **Status.**

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2.15 Debug Log window



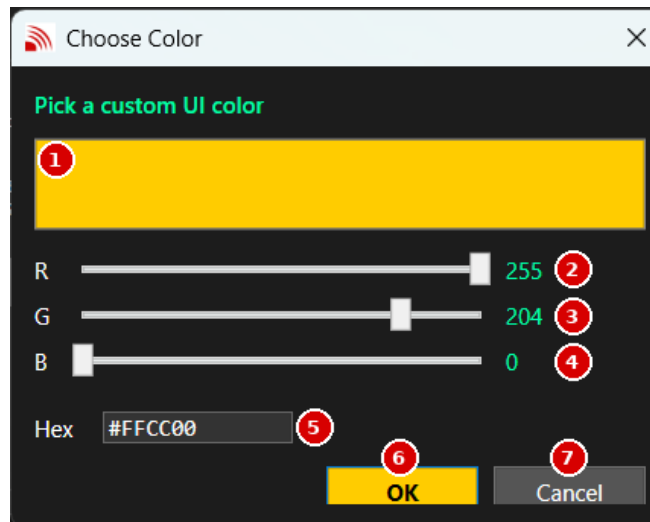
Debug Log window

1. **RESET LOGS.** Clears the display and starts a new log file.
2. **Pause.** Freezes the display; the log file keeps recording.
3. **EXIT.** Closes the window.
4. **Log list.**

Log files are in %LocalAppData%\MSCC-NET9\logs. Send them with any problem report.

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2.16 Choose Color window



Choose Color window

1. **Preview.**
2. **R** (0–255).
3. **G** (0–255).
4. **B** (0–255).
5. **Hex.** Type a color such as #FFD700 and press Enter.
6. **OK** (Enter).
7. **Cancel** (Esc).

2.17 Keyboard and mouse tips

- MSCC has no menus and no keyboard shortcuts. A mouse with a wheel is strongly recommended (as the 2022 guide said, especially on laptops).
- Wheel over a VFO digit tunes that digit. Wheel on the MAIN tab background tunes VFO A by Step. Wheel over a slider nudges it.
- Left-click the spectrum to tune. Right-click it to place the peak marker.
- Enter accepts the Host, Port and Hex boxes.



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□ **Appendix A: Updating the Radio Firmware (PSoC)**

The radio's main processor (a Cypress PSoC) holds two programs:

- the **application**, which is the radio firmware you update;
- a small **bootloader** ("LOADER") that you normally never touch.

Updating the application with a .cyacd file never changes the bootloader.

All radios must be updated to the latest firmware to use MSCC. Update the firmware before you run a new version of MSCC for the first time. After you connect, MSCC's title bar shows the firmware version the radio has now; compare it with the table below.

Which file: one per radio model, named for the radio and date, for example Proficio-MKII-PTT-20260822.cyacd. Current files by radio when this guide was written (firmware version shown in MSCC's title bar). If you were given a newer file, use it:

Radio	Firmware version
Proficio Legacy	1.231
Geminus MKII	2.151
Proficio MKII PTT	3.232
Proficio MKII ATU	4.151
Geminus Legacy	5.120
Ultimus Legacy	6.231
Ultimus MKII ATU	7.151
Ultimus MKII PTT	8.232

Where the files are: the firmware files come with the MSCC installer, in **C:\mscc-net9\PSoC Firmware**, one folder per radio.

You update the firmware over the radio's **USB cable** with the bootloader (A.1). The computer also needs the Multus USB driver, installed once with Zadig (A.2).

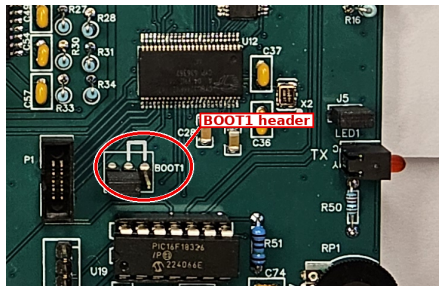
▲ Don't interrupt a firmware update. Once you click **Program**, don't unplug the USB cable, turn off the radio or close the program until the success message appears. It takes only a few seconds.

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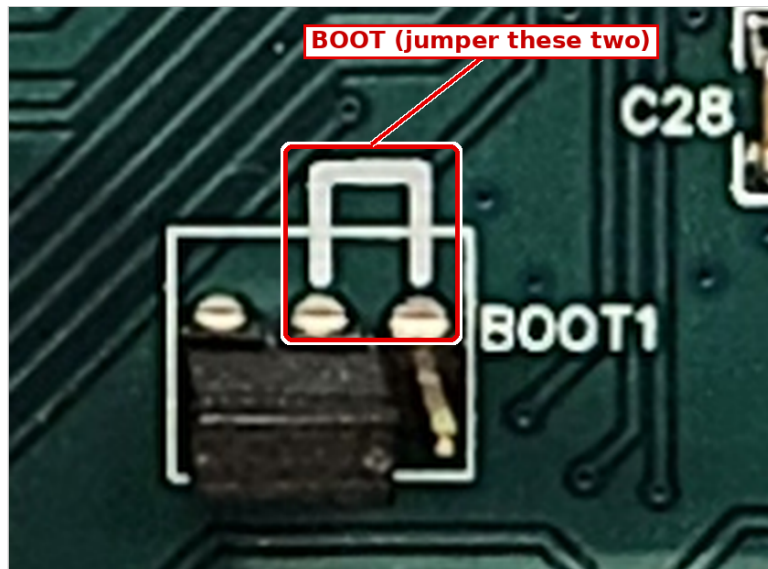
A.1 Updating the firmware over USB

You need the radio, its USB cable plugged into the computer, and the .cyacd file for your radio.

1. **Stop MSCC and the servers.**
2. **Power the radio off.** Always move the shunt (jumper) with the radio powered **off**. Leave the USB cable connected.
3. **Put the shunt on the boot pins.** Find the 3-pin **BOOT1** header (labeled on the board's silkscreen). A small “U” printed on the silkscreen joins two of its pins: those are the boot pins. With the radio still off, put the small plastic shunt (jumper) across the two pins joined by the printed U. The header's location differs by radio. The example below is a Geminus MKII board, where BOOT1 sits just right of connector P1 and above U19:



Example: Geminus MKII board, BOOT1 header circled

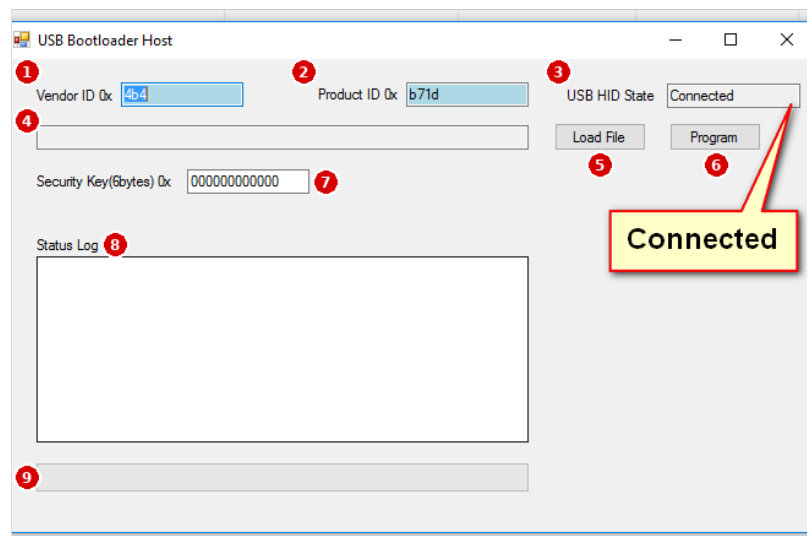


Close-up of BOOT1 on the Geminus MKII board: the printed U joins the two boot pins

Don't worry about the other pins. The other pins are just grounds. The radio doesn't need a shunt to operate, and a shunt on the wrong pins doesn't hurt anything.

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4. **Power the radio on.** The LED should blink **LOADER** in Morse code (very old firmware blinked “BOOT”). The radio is now ready to accept firmware. **If the LED isn't blinking**, the shunt is on the wrong pins: power off, move it to the two pins joined by the U, and try again.
5. **Start the upload program.** Run **bootloader.exe** in **C:\mscc-net9\utilities** (it needs **Bootloader_Utils.dll** and **CyUSB.dll**, which sit in the same folder). Its window is titled **USB Bootloader Host**.
6. **Wait for Connected.** The **USB HID State** box (3 in the picture) must show **Connected** before you go on. If it doesn't, check that the USB cable is plugged in and the LED is blinking **LOADER** (step 4).



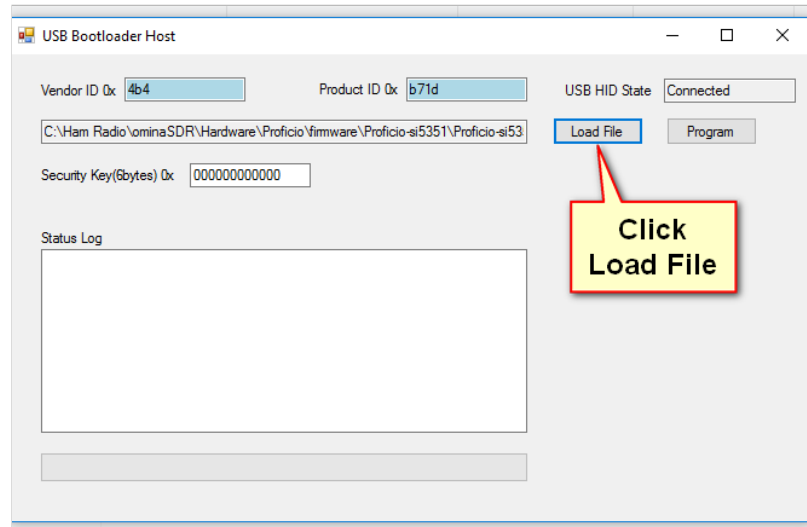
USB Bootloader Host, connected to the radio

1. **Vendor ID** (4b4). Leave as is.
 2. **Product ID** (b71d). Leave as is.
 3. **USB HID State.** Shows **Connected** when the radio is in boot mode and plugged in. Don't continue until it does.
 4. **File.** After Load File, shows the full folder path of the firmware file you picked.
 5. **Load File.** Choose the firmware file.
 6. **Program.** Starts the update.
 7. **Security Key.** Leave at all zeros.
 8. **Status Log.** Progress and the success message.
 9. **Progress bar.**
7. **Choose the firmware file.** Click **Load File**. In the window that opens, go to **C:\mscc-net9\PSoC Firmware** and open the folder for your radio, pick the **.cyacd** file for your radio, and click **Open**. The file's full path now shows in the long box under



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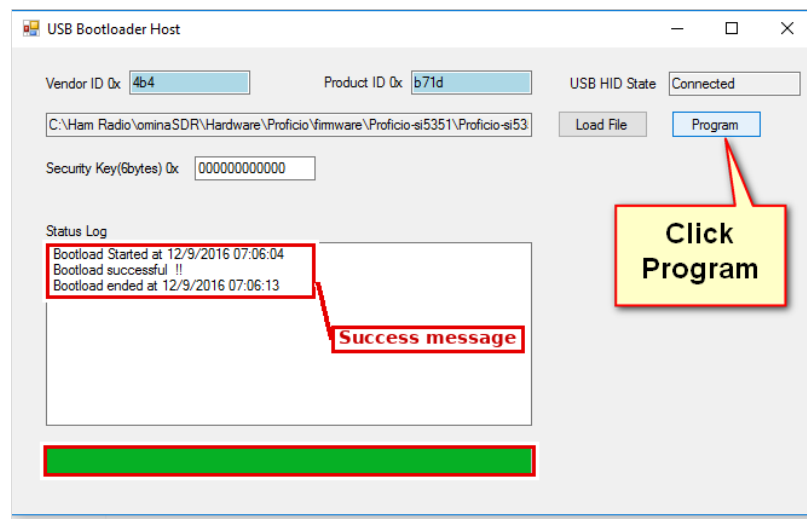
Vendor ID. (The path in the picture below is from an older install; yours will be in C:\mscc-net9.)



Click Load File and pick the .cyacd file for your radio

8. **Click Program.** In a few seconds the new firmware is loaded. **Don't interrupt the update.**

9. **Check for the success message.** The Status Log shows **Bootload successful !!** and the progress bar fills green.



Update finished: "Bootload successful !!" in the Status Log and a full green progress bar

10. **Take the shunt off the boot pins.** Power the radio **off** and remove the shunt from the two boot pins. You can leave it off, or park it on a single pin or on the other pins so it doesn't get lost; either is harmless. Then power the radio on again.

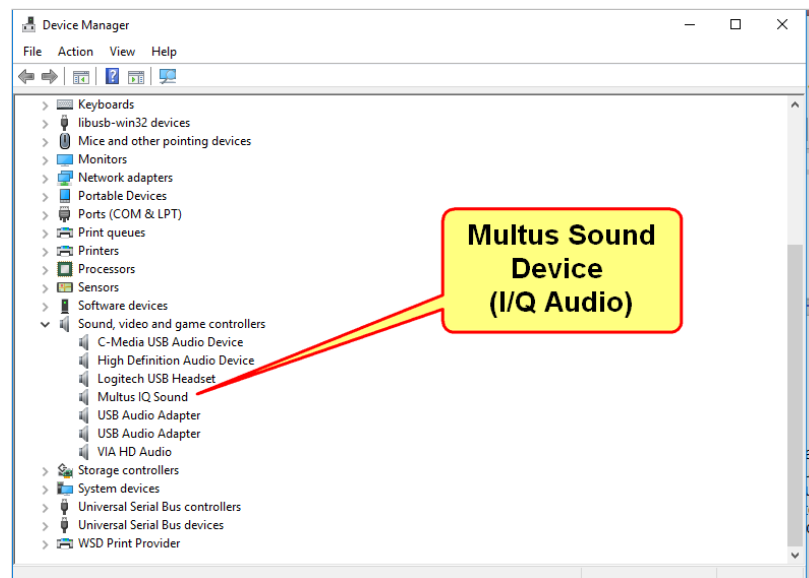
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11. **Check the version.** Close bootloader.exe, start the servers and MSCC, and connect. The title bar should show the new firmware version (see the table above).

A.2 Installing the USB driver (Zadig)

Zadig is used for all Multus radios now. Each Windows computer needs the **Multus SDR Control** driver, installed once with the current version of Zadig. Do this the first time you connect a Multus radio to a PC.

1. Plug the radio's USB cable into the PC and power cycle the radio.
2. The radio shows up in Windows **Device Manager** as a sound card named **Multus IQ Sound**. Windows installs it by itself because it is a standard USB audio device.

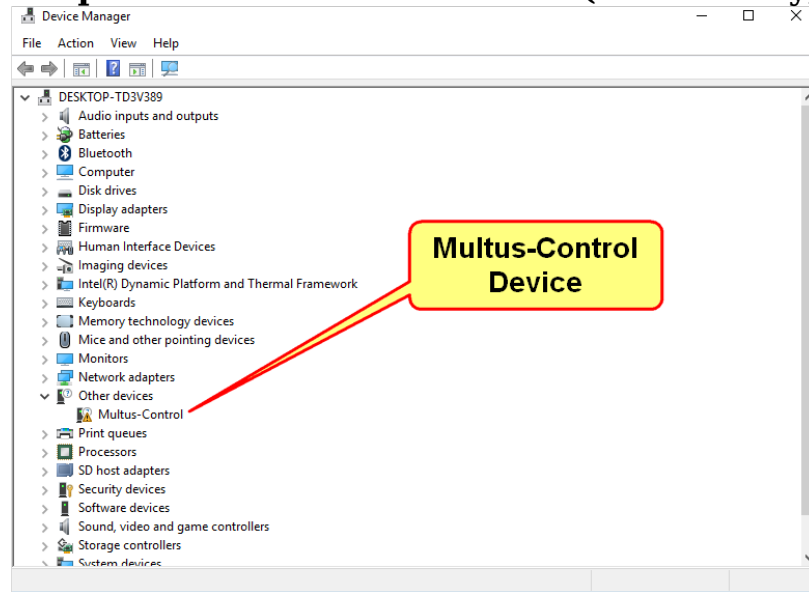


Device Manager: Multus IQ Sound under Sound, video and game controllers

⚠ Don't run Zadig on the Multus IQ Sound device.

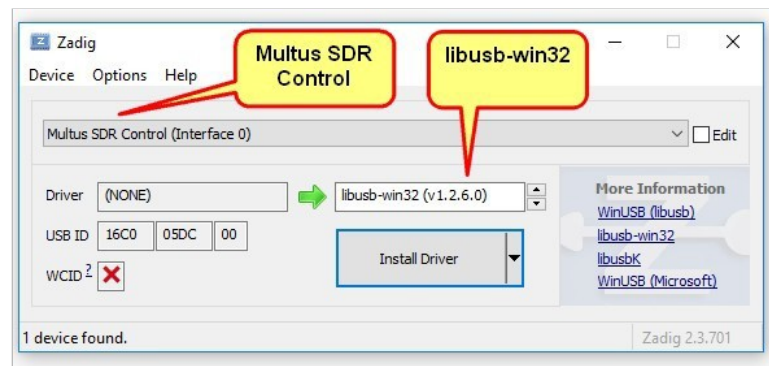
3. The radio also has a **Multus-Control** device (under **Other devices**) that needs a driver. Download the current version of **Zadig** from <https://zadig.akeo.ie/> and start it.

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Device Manager before the driver: Multus-Control under Other devices

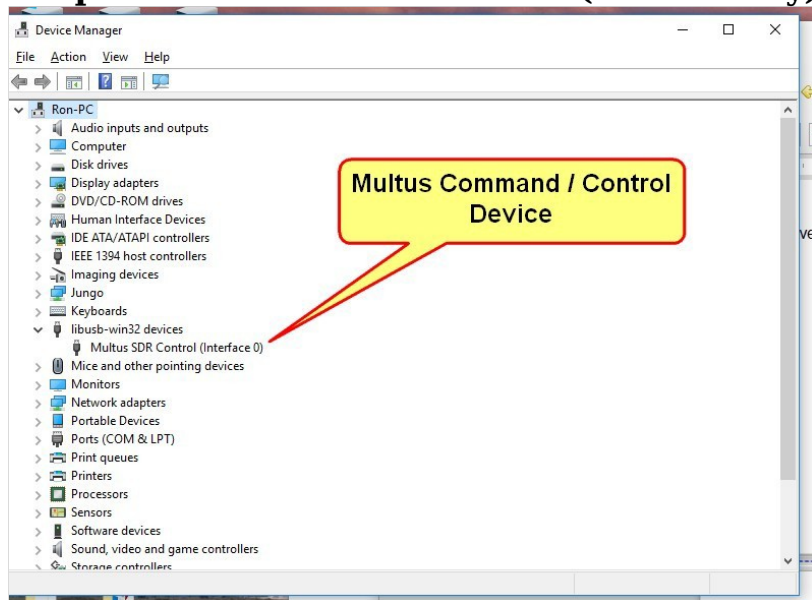
4. **Multus SDR Control** should show in Zadig's device box. If it isn't listed, click **Options** → **List All Devices**, then choose **Multus SDR Control** from the drop-down list. The driver box should show **libusb-win32**. Click **Install Driver**.



Zadig: Multus SDR Control, libusb-win32 driver, Install Driver button

5. If the driver installs, Zadig closes by itself. In Device Manager, the radio now shows as **Multus SDR Control (Interface 0)** under **libusb-win32 devices**. The driver installation is done.

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Device Manager after Zadig: Multus SDR Control under libusb-win32 devices